

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	19
Flg Calc [Int P]: Blind Flange	20
Flg Calc [Int P]: LHS Ch Flg	23
Internal Pressure Calculations:	32
Element: LHS Ch Shell	33
Element: Main Shell	35
Element: Dish End	37
Shop Pressure Test Results:	40
Field Pressure Test Results:	42
ASME TS Calc: Pressure Test	44
External Pressure Calculations:	52
Element and Detail Weights:	53
Nozzle Flange MAWP:	55
Wind Load Calculation:	57
Earthquake Load Calculation:	59
Center of Gravity Calculation:	61
Saddle Calcs: Road Transport:	62
Saddle Calcs: Operating Case:	76
Saddle Calcs: Test Case:	91
Saddle Calcs: Ext. Pressure:	105
Nozzle Calcs.: N1-Tube Inlet	118
Nozzle Calcs.: N2-Tube Outlet	129
Nozzle Calcs.: N3-Shell inlet	140
Nozzle Calcs.: N4-Shell Outlet	151
ASME TS Calc:	162



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\143-C.pvdb

Date of Analysis : Sep 9, 2026 9:36am

PV Elite 28, January 2026



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FileName : 143-C

Page 2 of 172

Vessel Design Summary:

Step: 29

9:36am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 930.000 mm ID
Vessel Design Length, Tangent to Tangent 7408.17 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 110 °C
Channel Side Design Temperature 82 °C
Shell Side Design Pressure 6400.000 KPa.
Channel Side Design Pressure 1000.000 KPa.
Shop Shell Side Test Pressure 9581.776 KPa.
Field Shell Side Test Pressure 8320.000 KPa.
Shop Channel Side Test Pressure 2182.259 KPa.
Field Channel Side Test Pressure 1300.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-266 2	...	...	K03506	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind flg	1013.115	103.42	1297.000	3.0000	N/A	No
LHS Ch Flg	1011.272	103.42	2725.000	3.0000	N/A	No
LHS Ch Shell	1008.949	103.42	4330.000	3.0000	N/A	No
Main Shell	6408.949	0.00	6578.000	3.0000	N/A	No
Dish End	6408.949	103.42	6502.000	3.0000	No	No

Liquid Level: 1338.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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FileName : 143-C

Page 3 of 172

Vessel Design Summary:

Step: 29 9:36am

Sep 9, 2026

Body Flg	83.2	80.0	88.0	80.0	74.0	...	1.00	1.00
Body Flg	204.2	121.0	93.0	85.0	77.9	...	1.00	1.00
Cylinder	1004.2	800.0	18.0	18.0	6.4	...	1.00	1.00
Cylinder	7358.2	6200.0	26.0	26.0	25.4	...	1.00	1.00
Ellipse	7408.2	50.0	30.0	25.0	24.7	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	80	66.0146	103.419	No Calc
20	30	85	77.9018	103.419	No Calc
30	Tsht	18	6.03432	103.419	2289
Tsht	50	18	No Calc	103.419	No Calc
50	60	26	No Calc	...	2199
60	70	25	5.53739	103.419	2808

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	920.969	32725.5	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	6327.53	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.899	Shell	Passed
N2-Tube Outlet	WRC-297	0.899	Shell	Passed
N3-Shell inlet	WRC-297	0.964	Shell	Passed
N4-Shell Outle	WRC-297	0.964	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	345.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1000.000 mm
Wear Pad Width	525.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1240.000 mm

Baseplate Length	980.000 mm
Baseplate Thickness	20.000 mm
Baseplate Width	360.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	500.000 mm
Number of Bolts in Baseplate	4



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FileName : 143-C

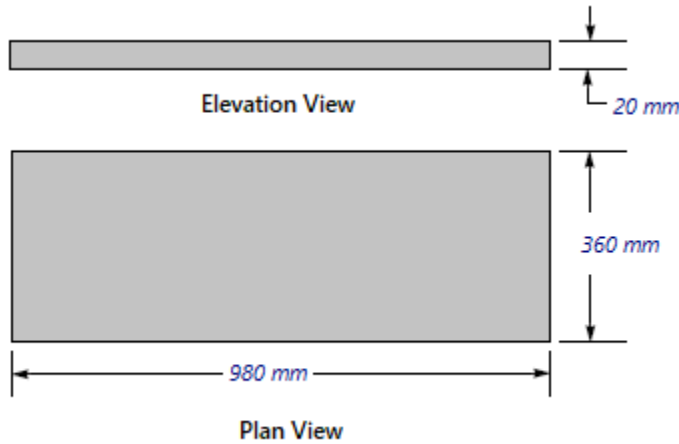
Page 4 of 172

Vessel Design Summary:

Step: 29

9:36am

Sep 9, 2026

Baseplate Sketch**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	14534.52	Kgf
Maximum Transverse Saddle Shear Load	836.36	Kgf
Maximum Longitudinal Saddle Shear Load	1042.99	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	12562.32	Kgf
Maximum Transverse Saddle Shear Load	110.11	Kgf
Maximum Longitudinal Saddle Shear Load	46.86	Kgf

Weights:

Fabricated - Bare w/o removable internals	13002.1	kg.
Shop Test - Fabricated + water (full)	17548.6	kg.
Shipping - Fab. + rem. intls.+ shipping app.	13002.1	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	13002.1	kg.
Empty - Fab. + intls. + details + wghts.	13632.3	kg.
Operating - Empty + operating liquid (no ca)	18175.8	kg.
Field Test - Empty weight + test medium (full)	17676.7	kg.

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FileName : 143-C

Page 5 of 172

Nozzle Summary:

Step: 27 9:36am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	3515	OK	...	OK	6.41	OK	Passed
N2-Tube Outlet	3506	OK	...	OK	6.44	OK	Passed
N3-Shell inlet	6450	...	...	OK	11.33	OK	Passed
N4-Shell Outlet	6441	...	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6441 Nozzle : N4-Shell Outlet [Shellside]
 Minimum MAWP Nozzles : 3506 Nozzle : N2-Tube Outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	604.175	90.000	711.999
30	N2-Tube Outlet	604.175	270.000	711.999
50	N3-Shell inlet	3158.175	90.000	611.999
50	N4-Shell Outlet	5358.174	270.000	611.999

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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FileName : 143-C

Page 6 of 172

Nozzle Schedule:

Step: 26 9:36am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3-Shell inlet	11.811 in	Actual	WNF	13.071	16.000	...	...	381.34	600
N4-Shell Outlet	11.811 in	Actual	WNF	13.071	16.000	...	...	381.34	600
N1-Tube Inlet	13.780 in	Actual	WNF	14.724	12.000	...	...	276.06	150
N2-Tube Outlet	13.780 in	Actual	WNF	14.724	12.000	...	...	276.06	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3-Shell in	SA-266 2	26.000	10.000	...	...	...
N4-Shell Ou	SA-266 2	26.000	10.000	...	...	...
N1-Tube Inl	SA-266 2	14.000	10.000	...	...	...
N2-Tube Out	SA-266 2	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3-Shell inlet	3158.175	90.0	300.00	0.00	Main Shell
N4-Shell Outlet	5358.174	270.0	300.00	0.00	Main Shell
N1-Tube Inlet	604.175	90.0	200.00	0.00	LHS Ch Shell
N2-Tube Outlet	604.175	270.0	200.00	0.00	LHS Ch Shell

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FileName : 143-C

Page 7 of 172

Bill of Material:

Step: 25 9:36am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 88.0mm THK X 1130.0mm OD	SA-266 2
1	WELD NECK FLANGE: 93.0mm THK X 1130.0mm OD	SA-266 2
1	CYLINDER: 18.0mm THK X 930.0mm ID X 800.0mm	SA-516 70
1	CYLINDER: 26.0mm THK X 930.0mm ID X 6200.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 30.0mm THK X 930.0mm ID X 50.0mm	SA-266 2
1	TUBESHEET: 982.0mm X 154.0mm THK	SA-266 2
482	TUBES: 5650.0mm X 25.4mm DIA X 2.1mm THK	SA-179
2	SADDLE: 345.0mm X 120 DEG	SA-516 70
1	INSULATION: 6200mm X 80mm THK	Hot Insulation
1	INSULATION: 50mm X 80mm THK	
1	GASKET: 988.0mm OD X 962.0mm ID	...
32	BODY FLANGE BOLTS: 28.6mm DIA	SA-193 B7
64	NUTS FOR BODY FLANGE BOLTS: 28.6mm DIA	...
1	NAMEPLATE	...

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FileName : 143-C

Page 8 of 172

MDMT Summary:

Step: 28 9:36am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell [8]	B	0	-1		0.977	26.000	1.00	No
Dish End [10]	B	-1	-2	-29	0.989	25.000	1.00	No
Dish End [7]	B	4	-5		0.832	30.000	1.00	No
Left Saddle [24]	B	-29	-30	-29	0.977	10.000	1.00	No
New Saddle [24]	B	-29	-30	-29	0.977	10.000	1.00	No
N3-Shell inlet[1]	B	0	-2		0.971	26.000	1.00	No
Nozzle Flg [5]	A	-18	-38					
N4-Shell Outle[1]	B	0	-1		0.972	26.000	1.00	No
Nozzle Flg [5]	A	-18	-38					
Tubesheet: SS[13]	B	10	-1		0.801	38.500	1.00	No
Warmest MDMT:		10	-1					
LHS Blind flg[11]	B	-7	-20	-29	0.769	20.000	1.00	No
LHS Ch Flg [11]	B	-11	-104	-29	0.299	18.000	1.00	No
LHS Ch Shell [8]	B	-11	-104	-29	0.295	18.000	1.00	No
N1-Tube Inlet [1]	B	-3	-104	-29	0.144	12.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
N2-Tube Outlet[1]	B	-3	-104	-29	0.145	12.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tubesheet: CS[14]	B	10	-104		0.147	38.500	1.00	No
Bolting [21]		-48						
Warmest MDMT:		10	-20					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-1.0	-12.0	Fail
Channel/Tube	-29.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.



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FileName : 143-C

Page 9 of 172

MDMT Summary:

Step: 28 9:36am Sep 9, 2026

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 143-C

Page 10 of 172

Warnings and Errors:

Step: 0 9:36am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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FileName : 143-C

Page 11 of 172

Input Echo:

Step: 1 9:36am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	6400	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	110.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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FileName : 143-C

Page 12 of 172

Input Echo:

Step: 1 9:36am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 13 of 172

Input Echo:

Step: 1 9:36am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Blind flg
Distance "FROM" to "TO"	80 mm
Flange Outside Diameter	1130 mm
Element Thickness	80 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	88 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1338 mm
Liquid Density	0.0009984 kg./cm ³

Element#	2/5
Element From Node	20



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FileName : 143-C

Page 14 of 172

Input Echo:

Step: 1 9:36am Sep 9,2026

Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	121 mm
Flange Inside Diameter	930 mm
Element Thickness	85 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	93 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1150 mm
Liquid Density	0.0009984 kg./cm ³

Element#	3/5
Element From Node	30
Element To Node	50
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	800 mm
Inside Diameter	930 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	910 mm
Liquid Density	0.0009984 kg./cm ³



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 15 of 172

Input Echo:

Step: 1 9:36am Sep 9,2026

Element From Node	30
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	13.78 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.82 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	30
Detail Type	Nozzle
Detail ID	N2-Tube Outlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	13.78 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.82 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	30
Detail Type	Weight
Detail ID	Partition plate
Dist. from "FROM" Node / Offset dist	400 mm
Miscellaneous Weight	300 Kg
Offset from Element Centerline	0 mm

Element#	4/5
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	6200 mm
Inside Diameter	930 mm
Element Thickness	26 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6400 KPa.
Design Temperature Internal Pressure	110 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	110 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1240 mm
Width of Saddle	345 mm
Height of Saddle at Bottom	1000 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	525 mm



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 16 of 172

Input Echo:

Step: 1 9:36am Sep 9,2026

Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.3000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Cumferential Eff. over Saddle	1.0	
Cumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	4960	mm
Width of Saddle	345	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	525	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Cumferential Eff. over Saddle	1.0	
Cumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	910	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	6200	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N3-Shell inlet	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	11.81	in.
Nozzle Schedule	60	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	194.3	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N4-Shell Outlet	
Dist. from "FROM" Node / Offset dist	4200	mm
Nozzle Diameter	11.81	in.
Nozzle Schedule	60	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	194.3	Kgf



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FileName : 143-C

Page 17 of 172

Input Echo:

Step: 1 9:36am Sep 9, 2026

Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2
Element From Node	50
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	1240 mm
Miscellaneous Weight	600 Kg
Offset from Element Centerline	990 mm
Element From Node	50
Detail Type	Weight
Detail ID	Wght:[2 of 2]
Dist. from "FROM" Node / Offset dist	4960 mm
Miscellaneous Weight	600 Kg
Offset from Element Centerline	990 mm

Element#	5/5
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	Dish End
Distance "FROM" to "TO"	50 mm
Inside Diameter	930 mm
Element Thickness	25 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6400 KPa.
Design Temperature Internal Pressure	110 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	110 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	225.4 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	910 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50.8 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³



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FileName : 143-C

Page 18 of 172

Input Echo:

Step: 1 9:36am Sep 9, 2026

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FileName : 143-C

Page 19 of 172

XY Coordinate Calculations:

Step: 2 9:36am

Sep 9, 2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind flg		83.175	...	80	...
LHS Ch Flg		204.175	...	121	...
LHS Ch Shell		1004.17	...	800	...
Main Shell		7358.17	...	6200	...
Dish End		7408.17	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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FileName : 143-C

Page 20 of 172

Flg Calc [Int P]: Blind Flange

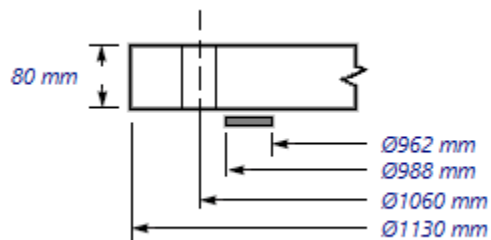
Flng: 5 9:36am Sep 9,2026

Flange Input Data Values

Description: Blind Flange :

LHS Blind flg

Flange Type	TEMA Channel Cover		
Design Pressure	P	1013.12	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1130.000	mm
Flange Thickness	t	80.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	3330.088	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	1.1625	mm
Diameter of Bolt Circle	C	1060.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	999.000	mm
Flange Face Inside Diameter	Fid	930.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	988.000	mm
Gasket Inside Diameter	Gi	962.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1150.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²





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FileName : 143-C

Page 21 of 172

Flg Calc [Int P]: Blind Flange Flng: 5 9:36am Sep 9,2026

Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width,	$b_o = N/2$	6.500 mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	975.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 975^2 \cdot 1013$$

$$= 77159.414 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 975 \cdot 3 \cdot 1013$$

$$+ 2 \cdot 1150 \cdot 5 \cdot 3 \cdot 1013$$

$$= 15764.655 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(77159 + 15765 + 0, 0)$$

$$= 92924.070 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 975 + 68.9 \cdot 5 \cdot 1150$$

$$= 178806.531 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((92924 + 0 + 4 \cdot 0 / 975) / 172, 178807 / 172)$$

$$= 101.951 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	101.951	150.296	
Radial Distance between Bolts and Edge:	28.575	35.000	
Circ. Spacing between the Bolts:	63.500	103.898	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 150 \cdot 172 / (68.9 \cdot \pi (988 + 962))$$

$$= 6.120 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 150$$

$$= 263597.16 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 92924.07 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1060 - 975) / 2$$

$$= 42.4245 \text{ mm}$$

Flange Stresses



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 22 of 172

Flg Calc [Int P]: Blind Flange Flng: 5 9:36am Sep 9, 2026

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 92924 \cdot 42.4 \cdot 1 / (77^2 \cdot 975) + 0.3 \cdot 1 \cdot 1013 \cdot 975^2 / 77^2$$

$$= 61.45 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 263597 \cdot 42.4 \cdot 1 / (77^2 \cdot 975)$$

$$= 36.04 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 92924 / 150$$

$$= 60.63 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (178807 / 150)$$

$$= 116.67 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	61.45	138.00	36.04	138.00
Bolting	60.63	172.00	116.67	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 1.0332 mm

Allowed deflection at center of Cover 1.1625 mm

Reqd. Blind Flange Thickness at Center 54.384 mm

Reqd. Blind Flange Thickness at Gasket 42.350 mm

Min. Required Flange Thickness per TEMA 74.033 mm

Estimated M.A.W.P. (Operating) 1310 KPa.

Estimated Finished Weight of Flange at given Thk. 621.8 kg.

Estimated Unfinished Weight of Forging at given Thk 621.8 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.77, Temperature Reduction per Fig. UCS 66.1 = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C

Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	56.93	138.00	33.39	138.00
Bolting	60.63	172.00	116.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 73.609 mm

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FileName : 143-C

Page 23 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am

Sep 9,2026

Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1011.27	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	930.000	mm
Flange Outside Diameter	A	1130.000	mm
Flange Thickness	t	85.0000	mm
Thickness of Hub at Small End	go	18.0000	mm
Thickness of Hub at Large End	gl	24.0000	mm
Length of Hub	h	36.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1060.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	999.000	mm
Flange Face Inside Diameter	Fid	930.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	988.000	mm
Gasket Inside Diameter	Gi	962.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1150.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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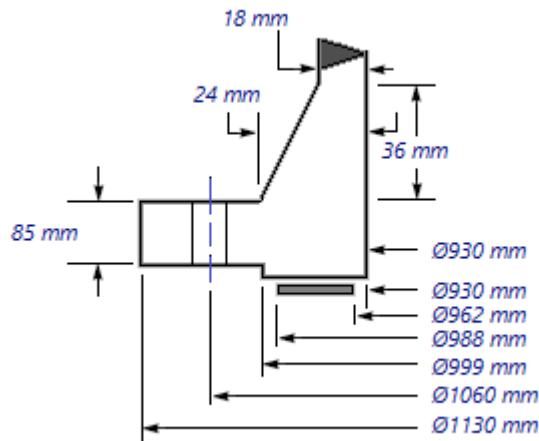
FileName : 143-C

Page 24 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am

Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1011 \cdot (930/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1011) + Ca$$

$$= 6.4449 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 15) / (468 + 0.6 \cdot 15)$$

$$= 4339.292 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18) / (465 + 0.6 \cdot 18)$$

$$= 5220.284 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	82.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	936.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	21.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	15.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	41.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	975.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 975^2 \cdot 1011$$

$$= 77019.070 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 975 \cdot 3 \cdot 1011$$

$$+ 2 \cdot 1150 \cdot 5 \cdot 3 \cdot 1011$$

$$= 15735.979 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 936^2 \cdot 1011$$

$$= 70958.805 \text{ Kgf}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 25 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am

Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= 77019 - 70959 \\
 &= 6060.266 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(77019 + 15736 + 0, 0) \\
 &= 92755.047 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.9 \cdot 6.42 \cdot \pi \cdot 975 + 68.9 \cdot 5 \cdot 1150 \\
 &= 178806.531 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a) \\
 &= \max((92755 + 0 + 4 \cdot 0 / 975) / 172, 178807 / 172) \\
 &= 101.951 \text{ cm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 28.6 + 6 \cdot 82 / (3 + 0.5) \\
 &= 197.721 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 1060 \cdot \sin(\pi / 32) \\
 &= 103.898 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ (Bs / (2a + t)) }, 1) \\
 &= \max(\sqrt{ (104 / (2 \cdot 28.6 + 82)) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	101.951	150.296	
Radial Distance between Hub and Bolts:	38.100	41.000	
Radial Distance between Bolts and Edge:	28.575	35.000	
Circ. Spacing between the Bolts:	63.500	103.898	197.721

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 150 \cdot 172 / (68.9 \cdot \pi (988 + 962)) \\
 &= 6.120 \text{ mm}
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172 \cdot 150 \\
 &= 263597.16 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 92755 - 77019 \\
 &= 15735.98 \text{ Kgf}
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 26 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am

Sep 9, 2026

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1060 - 975) / 2$$

$$= 42.4245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1060 - 936) / 2 + 42.4)$$

$$= 52.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1060 - 936 - 21) / 2$$

$$= 51.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	70959.	51.5000	1.0000	3654.
Face Pressure, Mt	6060.	52.2123	1.0000	316.
Gasket Load, Mg	15736.	42.4245	1.0000	668.
Gasket Seating, Matm	263597.	42.4245	1.0000	11183.

Total Moment for Operation, Mo 4638. Kg-m.
Total Moment for Gasket seating, Mg 11183. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 118.491 mm
Hub Ratio, h/h0 = HL / H0 0.304
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.400

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.885
Factor V per 4.16.5 0.391
Factor f per 4.16.5 1.010

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1130 / 936$$

$$= 1.207$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.21^2 \cdot \log_{10}(1.21))) / (1.21^2 - 1)) / (1.21 - 1)$$

$$= 10.413$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.21^2 (1 + 8.55246 \cdot \log_{10}(1.21)) - 1) / ((1.0472 + 1.9448 \cdot 1.21^2) \cdot (1.21 - 1))$$

$$= 1.836$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.21^2 (1 + 8.55246 \cdot \log_{10}(1.21)) - 1) / (1.36136 (1.21^2 - 1) \cdot (1.21 - 1))$$

$$= 11.443$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.21^2 + 1) / (1.21^2 - 1)$$

$$= 5.372$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 27 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am

Sep 9, 2026

$$= 11.4 \cdot 15^2 \cdot 118 / 0.39$$

$$= 780980.812 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.88 / 118$$

$$= 0.007 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (82 \cdot 0.0075 + 1) / 1.84 + 82^3 / 780981$$

$$= 1.584$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{12})$$

$$= (1.01 \cdot 4638 / 936) / (1.58 \cdot 21^2)$$

$$= 70.28 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{12})$$

$$= (1.01 \cdot 11183 / 936) / (1.58 \cdot 21^2)$$

$$= 169.45 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.82 \cdot 4638 / 936) / (1.58 \cdot 82^2)$$

$$= 8.29 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.82 \cdot 11183 / 936) / (1.58 \cdot 82^2)$$

$$= 19.98 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (10.4 \cdot 4638 / (82^2 \cdot 936)) - 5.37 \cdot 8.29$$

$$= 30.75 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (10.4 \cdot 11183 / (82^2 \cdot 936)) - 5.37 \cdot 20$$

$$= 74.13 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (70.3 + \max(8.29, 30.7)) / 2$$

$$= 50.51 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (169 + \max(20, 74.1)) / 2$$

$$= 121.79 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 92755 / 150$$

$$= 60.52 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (178807 / 150)$$

$$= 116.67 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 28 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am

Sep 9,2026

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	70.28	207.00	169.45	207.00
Radial Flange	8.29	138.00	19.98	138.00
Tangential Flange	30.75	138.00	74.13	138.00
Maximum Average	50.51	138.00	121.79	138.00
Bolting	60.52	172.00	116.67	172.00

Minimum Required Flange Thickness [Rigidity] 77.902 mm

Estimated M.A.W.P. (Operating) 2736 KPa.

Estimated Finished Weight of Flange at given Thk. 230.7 kg.

Estimated Unfinished Weight of Forging at given Thk 303.5 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	220.99	374.99	0.589	220.99	374.99	0.589
Tangential Flange	Sto	146.52	250.00	0.586	146.52	250.00	0.586
Combined	Sh/f + St	720.98	749.99	0.961	720.98	749.99	0.961
Combined	Sr + St	280.56	749.99	0.374	280.56	749.99	0.374
Combined	Sh + Sto	646.51	749.99	0.862	646.51	749.99	0.862
Radial Flange	Sr	59.57	499.99	0.119	59.57	499.99	0.119
Maximum Stress Ratio				1.000			1.000



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 29 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am Sep 9, 2026

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.26/1.26)

= 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/h0 < 0.35, else max (0.25; fn3)

= max(0.25; 0.36/1.22) if h/h0 < 0.35, else max (0.25; 0.24/0.77)

= 0.2930

In this case, the hub ratio h/h0, 0.3038 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	512.8 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 30 of 172

Flg Calc [Int P]: LHS Ch Flg Flng: 6 9:36am Sep 9,2026

Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.498 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (988^2 - 962^2) + (0.5 \cdot 1150 \cdot 5)$$

$$= 426.9 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208 \cdot 427 / (32 \cdot 4.7)$$

$$= 590.9 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(591; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 513)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 508 \geq 140 (427 / (32 \cdot 4.7))$$

$$:: 507.5 \geq 397.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 508 \geq (97 \cdot 427 + \pi/4 \cdot 1011 \cdot 962^2) / (0.7 \cdot 4.7 \cdot 32)$$

$$:: 507.5 \geq 463.5 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 508 \leq 275 (427 / (32 \cdot 4.7))$$

$$:: 507.5 \leq 781.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetaamax})$$

$$:: 508 \leq 513 (1/0.5)$$

$$:: 507.5 \leq 1029.5 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 508 \cdot 0.2 \cdot 4.7 \cdot 28.6$$

$$= 138.9 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 31 of 172

Flg Calc [Int P]: LHS Ch Flg

Flng: 6 9:36am

Sep 9, 2026

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 11183 / 1 \cdot 9807 \cdot 0.39 / (1.39 \cdot 201334 \cdot 15^2 \cdot 118 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 4638 / 1 \cdot 9807 \cdot 0.39 / (1.39 \cdot 197847 \cdot 15^2 \cdot 118 \cdot 0.3)$
 $= 0.422$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 11183 / 1 \cdot 9807 \cdot 0.39 / (1.58 \cdot 201334 \cdot 15^2 \cdot 118 \cdot 0.3)$
 $= 0.875$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 4638 / 1 \cdot 9807 \cdot 0.39 / (1.58 \cdot 197847 \cdot 15^2 \cdot 118 \cdot 0.3)$
 $= 0.369$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.3, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	61.54	207.00	144.76	207.00
Radial Flange	8.74	138.00	20.56	138.00
Tangential Flange	24.49	138.00	57.61	138.00
Maximum Average	43.02	138.00	101.18	138.00
Bolting	60.52	172.00	116.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 61.824 mm

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 32 of 172

Internal Pressure Calculations: Step: 5 9:36am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind flg		1013.1	88	3	1130	138	1
LHS Ch Flg		1011.3	93	3	930	138	2
LHS Ch Shell		1008.9	18	3	930	138	3
Main Shell		6408.9	26	3	930	138	4
Dish End		6408.9	30	3	930	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind flg		1000	1297	1687	80	74.033	1
LHS Ch Flg		1000	2725	2874	85	77.9018	2
LHS Ch Shell		1000	4330	5220	18	6.43699	3
Main Shell		6400	6578	7465	26	25.3594	4
Dish End		6400	6502	7379	25	24.6529	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 6441 KPa.
 Shell Side MAPnc = 7379 KPa.
 Channel Side MAWP = 1297 KPa.
 Channel Side MAPnc = 1687 KPa.

Elements Suitable for Design Internal Pressure.

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FileName : 143-C

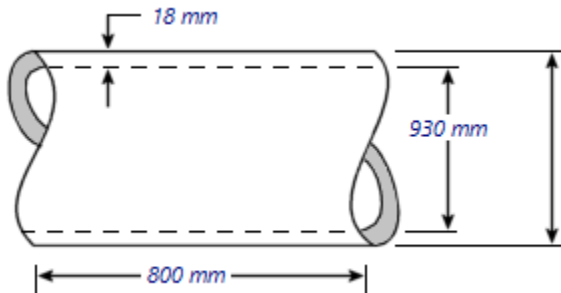
Page 33 of 172

Element: LHS Ch Shell

Step: 6 9:36am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1008.949 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	800.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	930.000 mm	Actual Stress	32.1 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	24278.2 cm ²	SG of Contents	1.000
Element Volume	543530.1 cm ³	Weight of Contents	540.3 Kg
Empty Weight	332.4 Kg	Operating Weight	872.7 Kg
		Comp. All. Ope. Stress	139.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1009 \cdot 468) / (138 \cdot 1 - 0.6 \cdot 1009) \\
 &= 3.4370 + 3.0000 = 6.4370 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.949 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 15) / (468 + 0.6 \cdot 15) \\
 &= 4339 - 8.95 = 4330 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (465 + 0.6 \cdot 18) \\
 &= 5220 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1009 \cdot (468 + 0.6 \cdot 15)) / (1 \cdot 15) \\
 &= 32.086 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 34 of 172

Element: LHS Ch Shell

Step: 6

9:36am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.899 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18$, $t_r = 4.42$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.29$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1296.790 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.295	Temperature Difference	93 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	966.00	920.97	64.40	0.9534	0.0027893	110.59

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 111) / (3 \cdot 64.4) = 2289 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.034	966.00	920.97	318.36	0.9534	0.0002470	24.70

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 24.7) / (3 \cdot 318) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	966.00	32725.54	64.40	33.8774	0.0002715	27.14

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 27.1) / (3 \cdot 64.4) = 562 \text{ KPa.}$$

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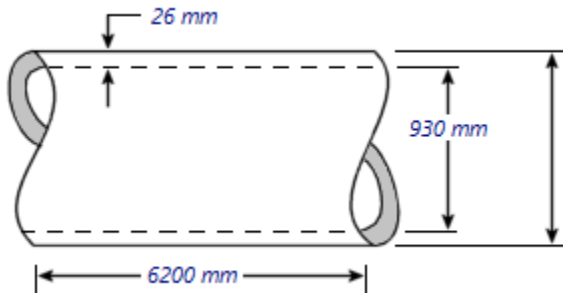
Page 35 of 172

Element: Main Shell

Step: 7 9:36am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	6408.949 KPa.	Int. Design Temperature	110 °C
Material	SA-516 70	Ext. Design Temperature	110 °C
Length	6200.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	930.000 mm	Actual Stress	134.3 N./mm ²
Spec. Min. Thk t,e, or sa	26.0000 mm	Spec. Nominal Thk.	26.000 mm
Surface Area	191272.7 cm ²	SG of Contents	1.000
Element Volume	2701013.2 cm ³	Weight of Contents	2699.4 Kg
Empty Weight	3752.3 Kg	Operating Weight	6451.7 Kg
		Comp. All. Ope. Stress	145.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6409 \cdot 468) / (138 \cdot 1 - 0.6 \cdot 6409)$$

$$= 22.3594 + 3.0000 = 25.3594 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.949 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 23) / (468 + 0.6 \cdot 23)$$

$$= 6587 - 8.95 = 6578 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (465 + 0.6 \cdot 26)$$

$$= 7465 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6409 \cdot (468 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 134.261 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 36 of 172

Element: Main Shell

Step: 7

9:36am

Sep 9, 2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 2.720 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 26$, $t_r = 22.5$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C

Min Metal Temp. at Required thickness (UCS 66.1) -1 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6441.336 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	26.000 mm	Basic MDMT	0 °C
Stress/Thk. Ratio	0.977	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-1 °C
		Joint Efficiency E^*	100 %

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FileName : 143-C

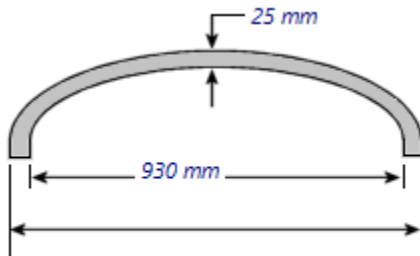
Page 37 of 172

Element: Dish End

Step: 8 9:36am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	6408.949 KPa.	Int. Design Temperature		110 °C
Material		SA-266 2	Ext. Design Temperature		110 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K03506
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	930.000 mm	Actual Stress		135.8 N./mm ²
Spec. Min. Thk t,e, or sa		25.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		12179.2 cm ²	SG of Contents		1.000
Element Volume		139279.6 cm ³	Weight of Contents		139.0 Kg
Empty Weight		291.5 Kg	Operating Weight		430.5 Kg
			Comp. All. Ope. Stress		145.3 N./mm ²
			Aspect Ratio		2.00

Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.812 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6409 \cdot 936 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6409)$$

$$= 21.6529 + 3.0000 = 24.6529 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.949 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 22) / (0.99 \cdot 936 + 0.2 \cdot 22)$$

$$= 6511 - 8.95 = 6502 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 25) / (1 \cdot 930 + 0.2 \cdot 25)$$

$$= 7379 \text{ KPa.}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 38 of 172

Element: Dish End

Step: 8 9:36am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6409 \cdot (0.99 \cdot 936 + 0.2 \cdot 22)) / (2 \cdot 1 \cdot 22)$$

$$= 135.830 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (6409 \cdot 468) / (138 \cdot 1 - 0.6 \cdot 6409) + 3$$

$$= 25.359 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 8.949 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 27) / (468 + 0.6 \cdot 27)$$

$$= 7695 - 8.95 = 7686 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (936 / (2 \cdot 236))^2) / 6$$

$$= 0.991534$$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 25, tr = 21.8, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.99, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 30, tr = 22.5, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.83, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-5 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6441.336 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	30.000 mm	Basic MDMT	4 °C
Stress/Thk. Ratio	0.832	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-5 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.625 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.812 mm

EXTERNAL PRESSURE RESULTS for: Dish End

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 110 °C

Elastic Modulus from Chart: CS-2 at 110 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure

MAEP



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 39 of 172

Element: Dish End

Step: 8

9:36am

Sep 9,2026

Tca	Outer Dia	Do/t	Factor A	Factor B
22.000	980.00	44.55	0.0031179	112.58

$$MAEP = B/(K0*Do/t) = 113/(0.9 *44.5) = 2808 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.537	980.00	386.22	0.0003596	35.95

$$MAEP = B/(K0*Do/t) = 36/(0.9 *386) = 103 \text{ KPa.}$$

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 40 of 172

Shop Pressure Test Results: Step: 9 9:36am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8373.736 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8320.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9581.776 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	7085.470 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	9152.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9152.000 KPa.
App 27-4	= M.A.W.P.	6441.336 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1685.827 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2182.259 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1426.469 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1296.790 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	69.7	234.0	0.298	2191.38
Main Shell	200.9	234.0	0.859	9590.89
Dish End	203.3	225.0	0.903	9590.89

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	138.00	1.000
N4-Shell Outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind flg	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 41 of 172

Shop Pressure Test Results: Step: 9 9:36am Sep 9,2026

Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	44.66	225.00	0.198
N2-Tube Outlet	44.66	225.00	0.198
N3-Shell inlet	118.64	225.00	0.527
N4-Shell Outlet	118.64	225.00	0.527

Elements Suitable for Test Pressure.

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 42 of 172

Field Pressure Test Results: Step: 10 9:36am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8373.736 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8320.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9581.776 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	7085.470 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	9152.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9152.000 KPa.
App 27-4	= M.A.W.P.	6441.336 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1685.827 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2182.259 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1426.469 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1296.790 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 6400 • 1

= 8320.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1000 • 1

= 1300.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	41.6	234.0	0.178	1309.12
Main Shell	174.5	234.0	0.746	8329.12
Dish End	176.5	225.0	0.785	8329.12

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	138.00	1.000
N4-Shell Outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind flg	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 43 of 172

Field Pressure Test Results: Step: 10 9:36am Sep 9, 2026

LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	26.68	225.00	0.119
N2-Tube Outlet	26.68	225.00	0.119
N3-Shell inlet	103.03	225.00	0.458
N4-Shell Outlet	103.03	225.00	0.458

Elements Suitable for Test Pressure.

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FileName : 143-C

Page 44 of 172

ASME TS Calc: Pressure Test

Case: 1 9:36a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: U Tube

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		8320.00 KPa.
Shell Thickness	ts	26.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	930.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	930.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	482
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5650.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5496.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	439.916 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	4723.3 cm ²
Perimeter of Tube Layout	Cp	2662.7 mm



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 45 of 172

ASME TS Calc: Pressure Test

Case: 1 9:36a Sep 9,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

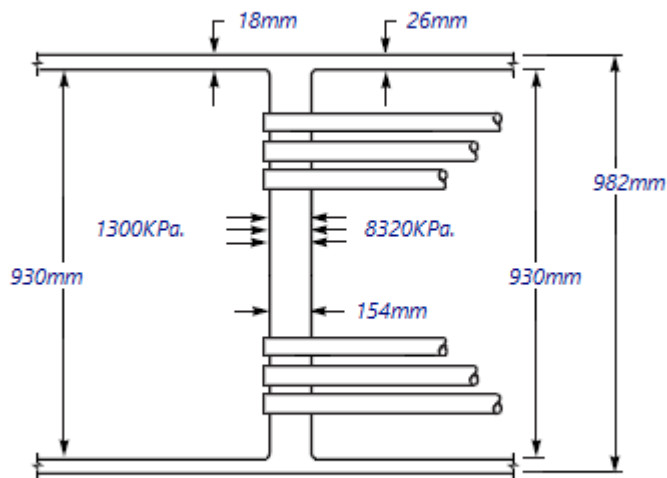
Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 154.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 982.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Channel Side Min. Design Pressure Ptd,min 103.4190 KPa.

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 46 of 172

ASME TS Calc: Pressure Test

Case: 1 9:36a Sep 9,2026

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1300 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 1300)$
 $= 0.1777 + 0.0000 = 0.1777 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5935.92	12.04	50.0000	0.0075908	95.82

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.8) / (3 \cdot 12) = 10612 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6872	25.40	5935.92	15.05	50.0000	0.0048534	95.13

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.1) / (3 \cdot 15.1) = 8425 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.6872 mm
Allowable Internal Pressure at Corroded thickness	16443.08 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	10612.12 KPa.
Required External Design Pressure	8423.42 KPa.
Required Thickness due to Shell Side pressure	1.6872 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 440 + 25.4$
 $= 905.232 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (31.8 - 25.4) / 31.8$
 $= 0.200$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 47 of 172

ASME TS Calc: Pressure Test

Case: 1 9:36a Sep 9, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 930/905$$

$$= 1.0274$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 930/905$$

$$= 1.0274$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/154$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.11 (202701232 / 202701232) \cdot$$

$$(92.4 / 138) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 905 \cdot 31.8) / (\pi \cdot 905^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 4.850394$; $\mu^* = 0.200000$; $E^* = 31489462 \text{ KPa}$.

$$E^*/E = 0.155349$$
 ; $v^* = 0.458572$

Note: As h/p (4.850) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (930 + 26) 26 }$$

$$= 0.0115 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - v_s^2))$$

$$= 0.012 \cdot 202701232 \cdot 26^3 / (6 (1 - 0.3^2))$$

$$= 7523568128.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (930 + 26) \cdot 7.52E+09 / 154^3 \cdot (1 + 154 \cdot 0.012 + (154 \cdot 0.012)^2 / 2)$$

$$= 51425100.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2 (D_s + t_s)))$$

$$= 930^2 / (4 \cdot 202701232 \cdot 26) \cdot (1 - 930 \cdot 0.3 / (2 (930 + 26)))$$

$$= 0.35039E-01 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 48 of 172

ASME TS Calc: Pressure Test

Case: 1

9:36a

Sep 9, 2026

$$= 1.03 \cdot 7.52E+09 \cdot 0.012 \cdot 0.035038803 (1 + 154 \cdot 0.012)$$

$$= 8668.1885 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(930 + 18)18}$$

$$= 0.0139 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.014 \cdot 202701232 \cdot 18^3 / (6(1 - 0.3^2))$$

$$= 3012981760.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(930 + 18 / \cos(0)) \cdot 3.01E+09 / 154^3 \cdot (1 + 154 \cdot 0.014 \cdot (1 + 1) / 2 + (154 \cdot 0.014)^2 / 2 \cdot 1)$$

$$= 25523956.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 930^2 / (4 \cdot 202701232 \cdot 18) \cdot (1 - 930 \cdot 0.3 / (2(930 + 18)))$$

$$= 0.50539E-01 \text{ mm/N} / \text{mm}^2$$

Channel coefficient [zeta α_c]:

$$zeta\alpha_c = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 3.01E+09 [0.050538629 (1 + 0.014 \cdot 1 \cdot 154) - 0.000 (1/154 + 1 \cdot 0.014/2)]$$

$$= 6842.8535 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 982/905$$

$$= 1.0848$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 31489462 \cdot (51425100 + 25523956 + 202701232 \cdot \ln(1.08))$$

$$= 1.6068$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot Pt]$$

$$= 905^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 8320 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 1300]$$

$$= 20219976.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= 20219976 + 6843 \cdot 1300 - 8668 \cdot 8320$$

$$= -43003640.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 49 of 172

ASME TS Calc: Pressure Test

Case: 1 9:36a Sep 9, 2026

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((-43003640) - 905^2/32 \cdot 1.61 (8320 - 1300)) / (1 + 1.61)$$

$$= -127301264.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= -127301264 + 905^2/64 \cdot (3 + 0.46) \cdot (8320 - 1300)$$

$$= 183565024.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| -127301264 | ; | 183565024 |)$$

$$= 183565024.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 183565024 / (0.2 (154 - 0)^2)$$

$$= 232.2175 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 250$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 166869392 / (0.2 (103 - 0)^2)$$

$$= 474.2835 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 103 + 0 + 0$$

$$= 102.7407 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(103 ; 32.8)$$

$$= 102.7407 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 8320 - 1300 |$$

$$= 7020.000 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.2 \cdot 154/905 \cdot 0.8 \cdot 0.95 \cdot 250$$

$$= 25.858 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 190) \cdot (4 \cdot 4723/2663) \cdot | 8320 - 1300 | + 0$$

$$= 32.7723 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 50 of 172

ASME TS Calc: Pressure Test

Case: 1 9:36a Sep 9, 2026

Note: For a given Shell thickness of: 26.000mm
Min. Shell len. adjacent to the tubesheet is: 279.899 mm

Note: For a given Channel thickness of: 18.000 mm
Min. Channel len. adjacent to the tubesheet is: 232.890mm

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (25.4 - 2.11) 2.11$
 $= 1.544 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 2.11 (25.4 - 2.11) 171$
 $= 2691.916 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
 $= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 171$
 $= 2634.3147 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / S_w$ where $S_w = \min(S \text{ and } S_t)$
 $= 171/171$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1$ for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
 $= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 171$
 $= 1704.5566 \text{ Kg}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$
 $= 2.667 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm
 Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 2692 is >= tube strength 2692 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	232.2 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	40.4 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	102.741	154.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 51 of 172

ASME TS Calc: Pressure Test

Case: 1 9:36a Sep 9, 2026

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	44.720	...	44	475	7	190 Ps+Pt D1 Ok
D2uc	114.322	...	280	475	49	190 Ps+Pt D2 Ok
D3uc	102.741	...	232	475	40	190 Ps+Pt D3 Ok
Max:	114.3216	... mm	0.590	0.255	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	370	...	370	Ok
D2uc	...	...	370	...	370	Ok
D3uc	...	...	370	...	370	Ok
Max	...	...	...	...	...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	114.322	154.000	Ok
Tube thickness :	1.687	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Given Shell Thickness: 26.0000 mm

Given Channel Thickness: 18.0000 mm

Min Shell length of thk, (26.000) adj. to tubesheet: 279.899 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 232.890 mm

Note: This is a full strength Tube to Tubesheet Joint.



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 52 of 172

External Pressure Calculations: Step: 11 9:36am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	77	No Calc	No Calc
20	30	No Calc	...	82	No Calc	No Calc
30	Tsht	920.969	966	15	0.0027893	110.586
Tsht	50	No Calc	...	15	No Calc	No Calc
50	60	6327.53	982	23	0.00070439	70.4229
60	70	No Calc	980	22	0.0031179	112.579

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	80	66.0146	103.419	No Calc
20	30	85	77.9018	103.419	No Calc
30	Tsht	18	6.03432	103.419	2289
Tsht	50	18	No Calc	103.419	No Calc
50	60	26	No Calc	...	2199
60	70	25	5.53739	103.419	2808

Minimum

2199

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	920.969	32725.5	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	6327.53	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 53 of 172

Element and Detail Weights:

Step: 12

9:36am

Sep 9, 2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	684.005	...	684.005	...	68.4005
20	30	250.744	88462.2	241.952	88778.8	25.0744
30	50	332.393	543530	277.871	550566	33.2393
50	60	3752.35	2701013	3329.8	2755541	375.235
60	70	291.531	139280	262.378	141771	29.1531
Total		5311	3472285.25	4796	3536656.75	531



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	...	40	-104	...	Liquid: 10
20	Liqd	88.4082	60.5	-110	...	Liquid: 20
30	Liqd	540.309	400	10	...	Liquid: 30
30	Nozl	109.802	400	640	...	N1-Tube Inlet
30	Nozl	109.802	400	-640	...	N2-Tube Outlet
30	Wght	300	400	...	...	Partition plate
50	Sadl	218.913	1240	714.5	...	Left Saddle
50	Sadl	218.913	4960	714.5	...	New Saddle
50	Liqd	2699.36	3100	10	...	Liquid: 50
50	Insl	583.186	3100	...	...	insulation
50	Nozl	213.727	2000	615	...	N3-Shell inlet
50	Nozl	213.727	4200	-615	...	N4-Shell Outlet
50	Wght	600	1240	-990	...	Fireproofing we
50	Wght	600	4960	-990	...	Wght:[2 of 2]
60	Liqd	139.004	127.5	10	...	Liquid: 60
60	Insl	47.0824	25.4	...	...	Insulation
30	FTsh	673.727	877	...	...	U Tube
30	Tube	3901.32	3625	...	...	Tubes

Total Weight of Each Detail Type:

Saddles	437.8
Liquid	3467.1
Insulation	630.3
Nozzles	647.1
Weights	1500.0
Exchanger Components	4575.0
Liquid in Tubes	1076.4

Sum of the Detail Weights 12333.7 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	5842.1	5842.1	5842.1	5842.1	5842.1	5842.1
Saddles	437.8	437.8	437.8	437.8	437.8	437.8
Nozzles	647.1	647.1	647.1	647.1	647.1	647.1



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 54 of 172

Element and Detail Weights:

Step: 12 9:36am

Sep 9, 2026

Wld Weights	1500.0	1500.0	1500.0	1500.0	1500.0	1500.0
Exchanger	4575.0	4575.0	4575.0	4575.0	4575.0	4575.0
Insulation	...	...	...	...	630.3	630.3 *
Ope. Liquid	...	...	...	...	...	3467.1
Tube Ope Lqd	...	...	...	...	...	1076.4
Test Liquid	...	3470.2	...	...	...	...
Tube Tst Lqd	...	1076.4	...	...	...	...
Totals	13002.1	17548.6	13002.1	13002.1	13632.3	18175.8

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	13002.1 kg.
Shop Test Wt.	- Fabricated weight + water (full)	17548.6 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	13002.1 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	13002.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	13632.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	18175.8 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	17609.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	17676.7 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	1077024.6 cm ³
Weight of Ope Liq in tubes :	1076.4 kg.
Weight of Water in tubes :	1076.4 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	13152.7	...
20	30	7643.43	3768.97
30	50	24278.2	23373.5
50	60	191273	181144
60	70	12179.2	10836.2

Total	248526.4	219122.9 cm ²
Total	267.5	235.9 ft ²

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 55 of 172

Nozzle Flange MAWP:

Step: 13 9:36am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	1820.4	1960.0	82	150	GR 1.1	2318.5	1686.4	1820.4	412.1
N2-Tube Outlet	1820.4	1960.0	82	150	GR 1.1	2318.5	1686.4	1820.4	412.1
N3-Shell inlet	9274.1	10200.0	110	600	GR 1.1	2332.2	9274.1	9274.1	9274.1
N4-Shell Outle	9274.1	10200.0	110	600	GR 1.1	4075.0	9274.1	9274.1	9274.1

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 9274.069 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 10200.003 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1686.430 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	389.001	1.20	2472	2677	1296.8	1686.4	Pass
N2-Tube Outlet	389.001	1.20	2472	2677	1296.8	1686.4	Pass
N3-Shell inlet	350.901	0.50	1816	2294	6441.3	9274.1	Pass
N4-Shell Outlet	350.901	0.50	3244	3212	6441.3	9274.1	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2472) + 4 \cdot \text{abs}(2677) \cdot 389.001 \leq \pi \cdot 389.001^3 [(1820.434 - 1296.790) + 1.200 \cdot 1820.434]$$

$$= 0.43724E+05 \leq 0.51072E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 43723.9 / (\pi \cdot 389.001^3))$$

$$= \min(1820.434, 1686.430)$$

$$= 1686.430 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2472) + 4 \cdot \text{abs}(2677) \cdot 389.001 \leq \pi \cdot 389.001^3 [(1820.434 - 1296.790) + 1.200 \cdot 1820.434]$$

$$= 0.43724E+05 \leq 0.51072E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 43723.9 / (\pi \cdot 389.001^3))$$

$$= \min(1820.434, 1686.430)$$

$$= 1686.430 \text{ KPa.}$$

Evaluating Flange on Nozzle [N3-Shell inlet]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 56 of 172

Nozzle Flange MAWP:

Step: 13 9:36am Sep 9, 2026

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1816) + 4 \cdot \text{abs}(2294) 350.901 \leq \pi \cdot 350.901^3 [(9274.069 - 6441.336) + .500 \cdot 9274.069] \\ &= 0.32284E+05 \leq 0.10340E+06 \quad \text{Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3-Shell inlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9274.069, 0.500 \cdot 9274.069 + 9274.069 - 32283.7 / (\pi \cdot 350.901^3)) \\ &= \min(9274.069, 11578.855) \\ &= 9274.069 \text{ KPa.} \end{aligned}$$

Evaluating Flange on Nozzle [N4-Shell Outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(3244) + 4 \cdot \text{abs}(3212) 350.901 \leq \pi \cdot 350.901^3 [(9274.069 - 6441.336) + .500 \cdot 9274.069] \\ &= 0.56408E+05 \leq 0.10340E+06 \quad \text{Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell Outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9274.069, 0.500 \cdot 9274.069 + 9274.069 - 56407.7 / (\pi \cdot 350.901^3)) \\ &= \min(9274.069, 9836.090) \\ &= 9274.069 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 57 of 172

Wind Load Calculation:

Step: 14 9:36am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.92)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.601
Structure Height to Diameter ratio	7.129

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1000 + 300 = 1300 \text{ mm}$$

$$= 4.27 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (4.27 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Z_g)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 58 of 172

Wind Load Calculation:

Step: 14

9:36am

Sep 9, 2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.6 • 1.17

= 9.54 lbs. [4.33] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind flg	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Flg	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Shell	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Dish End	1300.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition pla	0.00	1300.00	78.12	0.00
Fireproofing	0.00	300.00	78.12	0.00
Wght:[2 of 2]	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	1356	1084.8	78.12	4.75987
20	30	1300	1116	1350.36	78.12	5.9251
30	50	1300	1159.2	9273.6	78.12	40.6906
50	60	1300	1370.4	84964.8	78.12	372.808
60	70	1300	1368	3705.82	78.12	16.2604

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 59 of 172

Earthquake Load Calculation: Step: 15 9:36am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(5.130.75)$
 $= 0.068$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.068, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 17738 \cdot 1.25$
 $= 1042.991 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 17738$
 $= 556.3 \text{ Kgf}$

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	565	2533.99	175.645



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 60 of 172

Earthquake Load Calculation:

Step: 15

9:36am

Sep 9, 2026

20	30	465	2533.99	144.558
30	50	465	2533.99	144.558
50	Sad1	465	2533.99	144.558
Sad1	60	465	2533.99	144.558
50	60	465	2533.99	144.558
60	70	465	2533.99	144.558

The Earthquake Weight does not include the weight of the saddles.

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FileName : 143-C

Page 61 of 172

Center of Gravity Calculation: Step: 16 9:36am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4258.174 mm
Center of Gravity of the Liquid	3713.219 mm
Center of Gravity of the Insulation	4491.648 mm
Center of Gravity of the Nozzles	3018.048 mm
Center of Gravity of the Added Weights (Operating)	3527.375 mm
Center of Gravity of the Added Weights (Empty)	3527.375 mm
Center of Gravity of the Tubesheet(s)	1081.175 mm
Center of Gravity of the Tubes	3829.175 mm
Center of Gravity of Bare Shell New and Cold	3470.234 mm
Center of Gravity of Bare Shell Corroded	3415.009 mm
Vessel CG in the Operating Condition	3555.990 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3464.763 mm
Vessel CG in the Test Condition	3535.875 mm

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 62 of 172

Saddle Calcs: Road Transport:

Step: 17

9:36am

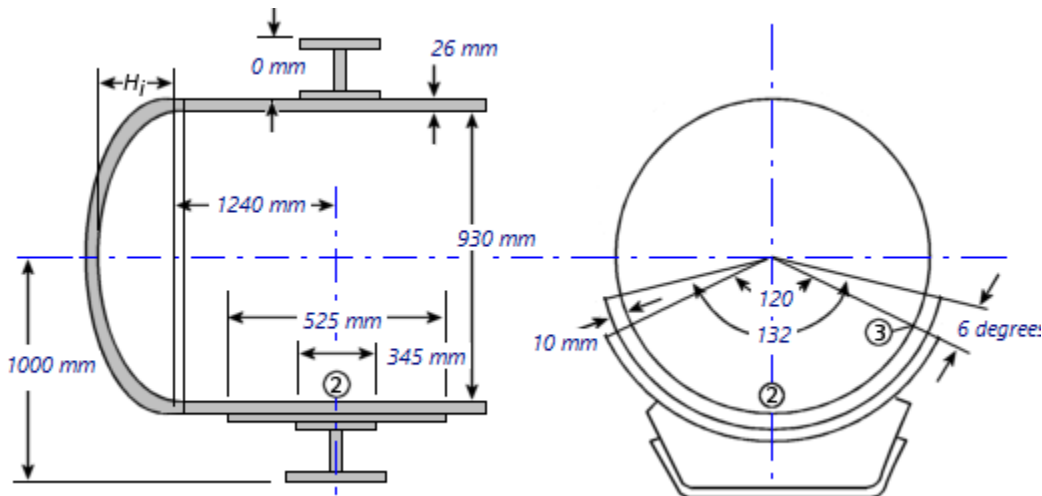
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	478.00	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		10471.81	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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FileName : 143-C

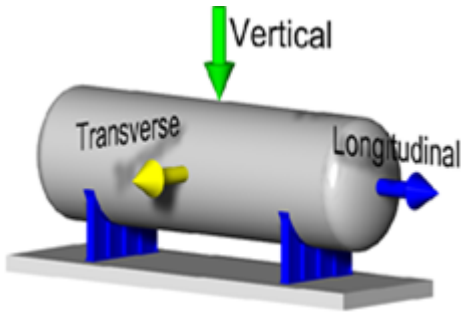
Page 63 of 172

Saddle Calcs: Road Transport:

Step: 17

9:36am

Sep 9, 2026



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-1.88	-234.00
Long. Stress at Top of Midspan	1.88	138.00
Long. Stress at Bottom of Midspan	1.88	138.00
Long. Stress at Top of Saddles	11.75	138.00
Long. Stress at Bottom of Saddles	-6.52	-234.00
Long. Stress at Bottom of Saddles	6.52	138.00
Tangential Shear in Shell	5.84	110.40
Circ. Stress at Horn of Saddle	7.66	207.00
Circ. Compressive Stress in Shell	0.42	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 108 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 917.8 \text{ lbf} \\
 &= 416.306 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 3.22^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 83.1 \text{ lbf} \\
 &= 37.7 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 12564 \cdot 1 + 416 \\
 &= 1672.7 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 12564 \cdot 1 + 37.7 \\
 &= 1294.1 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 1673/2 \cdot 1000/868 \\
 &= 963.8 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 1294 \cdot 1000/3720 \\
 &= 347.9 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 9055 \cdot 0.05
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 64 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9, 2026

= 452.761 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 9055 + 453 + max(348, 964)

= 10471.8 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 10690.72 Kgf

Transverse Shear Load 836.36 Kgf

Longitudinal Shear Load 1294.12 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(930/2 + 26 + 10)sin(120/2), 2•478)

= 867.758 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -10472•1240[1-(1-1240/6250+(478²-0²)/

(2•1240•6250))/(1+(4•0)/(3•6250))]

= -2384.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 10472•6250/4((1+2(478²-0²)/(6250²))/(1+(4•0)/

(3•6250))-4•1240/6250)

= 3568.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 478/(2•26) - 3569/(π•478²•26)

= -1.88 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 478/(2 • 26) + 3569/(π • 478² • 26)

= 1.88 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•478/(2•26)-2385/(0.11•π•478²•26)

= 11.75 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•478/(2•26)+2385/(0.19•π•478²•26)

= -6.52 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 10472 (6250 - 2 • 1240)/(6250 + (4 • 0/3))

= 6316.6 Kgf



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 65 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9,2026

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 6317 / (478 \cdot 26)$$

$$= 5.84 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(478 \cdot 26)}$$

$$= 86.955 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(345 + 1.56 \cdot \sqrt{(478 \cdot 26)}, 2 \cdot 1240)$$

$$= 518.91 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 10472 \cdot 0.1 / (519(26 + 1 \cdot 10))$$

$$= -0.42 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -10472 / (4(26 + 1 \cdot 10)519) -$$

$$3 \cdot 0.053 \cdot 10472 / (2(26 + 1 \cdot 10)^2)$$

$$= -7.66 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(465 \cdot 26)} + 525)26^3) / 12$$

$$= 102.018 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 102 + 181(156 - 13)^2$$

$$= 36889.758 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	696.5	26.0	13.0	181.1	235426.8	0.369E+05
Wearplate	525.0	10.0	31.0	52.5	162750.0	0.815E+04
Web	10.0	479.0	275.5	47.9	1319645.2	0.161E+05
BasePlate	360.0	20.0	525.0	72.0	3780000.0	0.983E+05
Totals	...	...	...	353.5	5497822.0	0.159E+06

Distance to Centroid [C1]:

$$= AY / A$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 66 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9, 2026

$$= 5497822/353$$

$$= 155.526 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 10472$$

$$= 2131.2407 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.2123 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 465 \cdot \sin(120/2) / 1.05$$

$$= 615.448 \text{ mm}$$

Bending Moment, M = Fh * d = 1311.6938 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.2551 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(10472 + 219)360 / (4 \cdot 980 \cdot 173))}$$

$$= 12.909 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (465 + 26 + 10)$$

$$= 867.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 868 / (3 - 1)$$

$$= 433.879 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 434 \cdot 360 / 2$$

$$= 780.982 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 10691 / (980 \cdot 360)$$

$$= 3.030 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 781 \cdot 3.03$$

$$= 2366.571 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 67 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9,2026

$$= 33.5 + 21.7$$

$$= 55.194 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 2367/55.2$$

$$= 4.205 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 345 - 10$$

$$= 335.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/55.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345/2$$

$$= 172.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3422/55.2)}$$

$$= 78.739 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (465 + 26 + 10) - 20$$

$$= 729.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 730/78.7$$

$$= 9.265$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1294 / (2 \cdot 980) \cdot 434 \cdot 730/2$$

$$= 104.494 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.26 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.26)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (9.26) / (8 \cdot 123) - (9.26^3) / (8 \cdot 123^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.2/153 + (104 \cdot 172/34219688) / 173$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 68 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9, 2026

= 0.057

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/76.9

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 345/2

= 172.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1000 - cos(501/2)(20 + 0 + 0) - 0

= 479.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{3425/76.9}$

= 66.747 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 479/66.7

= 7.176

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 1294/(2 • 980)

= 0.660 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.66 • 434 • 479

= 137.224 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 137/198580

= 6.777 N./mm²

Compressive Allowable, KL/r < Cc (7.18 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (7.18)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (7.18) / (8 \cdot 123) - (7.18^3) / (8 \cdot 123^3))$

= 153.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 6.12/154 + (137 • 172/3425)/173

= 0.079

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 69 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9,2026

Vessel Mean Radius	Rm	478.00	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	245.00	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		4648.27	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.63	-234.00
Long. Stress at Top of Midspan	0.63	138.00
Long. Stress at Bottom of Midspan	0.63	138.00
Long. Stress at Top of Saddles	6.47	138.00
Long. Stress at Bottom of Saddles	-3.59	-234.00
Long. Stress at Bottom of Saddles	3.59	138.00
Tangential Shear in Shell	2.46	110.40
Circ. Stress at Horn of Saddle	3.40	207.00
Circ. Compressive Stress in Shell	0.19	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 108 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 917.8 \text{ lbf} \\
 &= 416.306 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 3.22^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 83.1 \text{ lbf} \\
 &= 37.7 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 70 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9, 2026

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 12564 \cdot 1 + 416 \\
 &= 1672.7 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [FI]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 12564 \cdot 1 + 37.7 \\
 &= 1294.1 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 1673/2 \cdot 1000/868 \\
 &= 963.8 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= FI \cdot B / Ls \\
 &= 1294 \cdot 1000/3720 \\
 &= 347.9 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 3509 \cdot 0.05 \\
 &= 175.450 \text{ Kgf}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 3509 + 175 + \max(348, 964) \\
 &= 4648.3 \text{ Kgf}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4867.19	Kgf
Transverse Shear Load	836.36	Kgf
Longitudinal Shear Load	1294.12	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm) \\
 &= \min(2(930/2 + 26 + 10)\sin(120/2), 2 \cdot 478) \\
 &= 867.758 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -4648 \cdot 1240 [1 - (1 - 1240/6250 + (478^2 - 245^2)/(2 \cdot 1240 \cdot 6250))/(1 + (4 \cdot 245)/(3 \cdot 6250))] \\
 &= -1313.5 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 4648 \cdot 6250/4((1 + 2(478^2 - 245^2)/(6250^2))/(1 + (4 \cdot 245)/(3 \cdot 6250)) - 4 \cdot 1240/6250) \\
 &= 1197.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 71 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9, 2026

$$= 0 \cdot 478 / (2 \cdot 26) - 1198 / (\pi \cdot 478^2 \cdot 26)$$

$$= -0.63 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 478 / (2 \cdot 26) + 1198 / (\pi \cdot 478^2 \cdot 26)$$

$$= 0.63 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 478 / (2 \cdot 26) - 1314 / (0.11 \cdot \pi \cdot 478^2 \cdot 26)$$

$$= 6.47 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 478 / (2 \cdot 26) + 1314 / (0.19 \cdot \pi \cdot 478^2 \cdot 26)$$

$$= -3.59 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 4648 (6250 - 2 \cdot 1240) / (6250 + (4 \cdot 245 / 3))$$

$$= 2664.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 2665 / (478 \cdot 26)$$

$$= 2.46 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(478 \cdot 26)}$$

$$= 86.955 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(345 + 1.56 \cdot \sqrt{(478 \cdot 26)}, 2 \cdot 1240)$$

$$= 518.91 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 4648 \cdot 0.1 / (519(26 + 1 \cdot 10))$$

$$= -0.19 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -4648 / (4(26 + 1 \cdot 10) 519) -$$

$$3 \cdot 0.053 \cdot 4648 / (2(26 + 1 \cdot 10)^2)$$

$$= -3.40 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 72 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9,2026

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(465 \cdot 26) + 525})^3) / 12$$

$$= 102.018 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 102 + 181 (156 - 13)^2$$

$$= 36889.758 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	696.5	26.0	13.0	181.1	235426.8	0.369E+05
Wearplate	525.0	10.0	31.0	52.5	162750.0	0.815E+04
Web	10.0	479.0	275.5	47.9	1319645.2	0.161E+05
BasePlate	360.0	20.0	525.0	72.0	3780000.0	0.983E+05
Totals	...	...	...	353.5	5497822.0	0.159E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5497822 / 353$$

$$= 155.526 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 4648$$

$$= 946.0245 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5381 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000 - 465 \cdot \sin(120 / 2) / 1.05$$

$$= 615.448 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 582.2405 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.5571 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4648 + 219) \cdot 360) / (4 \cdot 980 \cdot 173)}$$

$$= 8.710 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (465 + 26 + 10)$$

$$= 867.758 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 73 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9,2026

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 868 / (3 - 1)$$

$$= 433.879 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 434 \cdot 360 / 2$$

$$= 780.982 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4867 / (980 \cdot 360)$$

$$= 1.380 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 781 \cdot 1.38$$

$$= 1077.433 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 33.5 + 21.7$$

$$= 55.194 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1077 / 55.2$$

$$= 1.914 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 345 - 10$$

$$= 335.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 55.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345 / 2$$

$$= 172.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (3422 / 55.2) }$$

$$= 78.739 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120 / 2) (465 + 26 + 10) - 20$$

$$= 729.500 \text{ mm}$$

Intermediate Term [Cc]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 74 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9,2026

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 730 / 78.7$$

$$= 9.265$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1294 / (2 \cdot 980) \cdot 434 \cdot 730 / 2$$

$$= 104.494 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.26 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.26)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (9.26) / (8 \cdot 123) - (9.26^3) / (8 \cdot 123^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.91 / 153 + (104 \cdot 172 / 34219688) / 173$$

$$= 0.042$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 76.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345 / 2$$

$$= 172.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000 - \cos(501/2) (20 + 0 + 0) - 0$$

$$= 479.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3425 / 76.9)}$$

$$= 66.747 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 479 / 66.7$$

$$= 7.176$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1294 / (2 \cdot 980)$$

$$= 0.660 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 75 of 172

Saddle Calcs: Road Transport: Step: 17 9:36am Sep 9, 2026

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.66 \cdot 434 \cdot 479 \\
 &= 137.224 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 137 / 198580 \\
 &= 6.777 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($7.18 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (7.18)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (7.18) / (8 \cdot 123) - (7.18^3) / (8 \cdot 123^3)) \\
 &= 153.7 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba} \\
 &= 2.79 / 154 + (137 \cdot 172 / 3425) / 173 \\
 &= 0.057
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 76 of 172

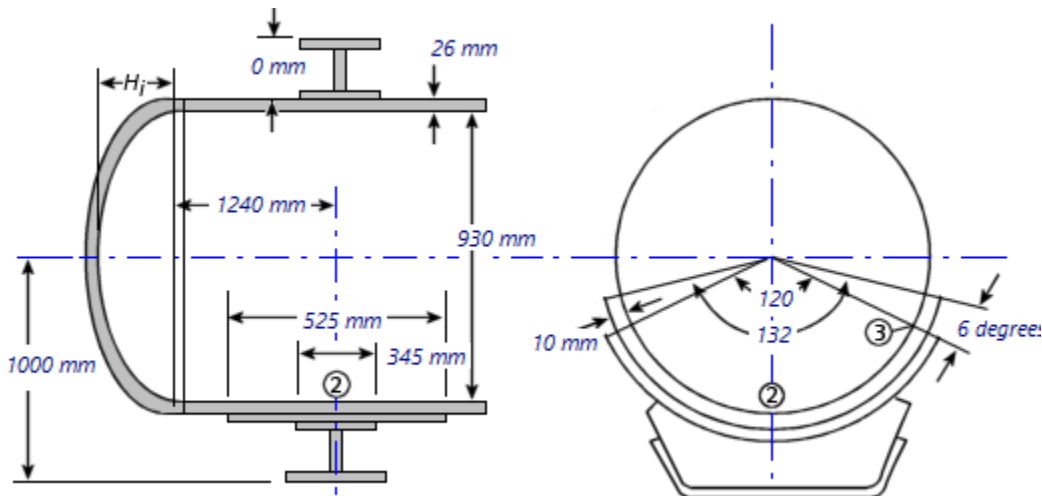
Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	479.50	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	14315.60	Kgf
Pressure used in Analysis	P	6404.362 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	63.88	138.00
Long. Stress at Bottom of Midspan	69.64	138.00
Long. Stress at Top of Saddles	84.80	138.00
Long. Stress at Bottom of Saddles	56.76	138.00
Tangential Shear in Shell	8.99	110.40
Circ. Stress at Horn of Saddle	12.31	172.50
Circ. Compressive Stress in Shell	0.64	138.00



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 77 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (440/2 + 0) \cdot 1000/868$$

$$= 761.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(93.7, 0) \cdot 1000/3720$$

$$= 25.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(1043, 0) \cdot 1000/3720$$

$$= 280.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1043/2 + 0) \cdot 1000/868$$

$$= 1802.9 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 12513 + \max(25.2, 761, 280, 1803)$$

$$= 14315.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(0.98 \cdot 1.2)^2)$$

$$= 919.063 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	14534.52	Kgf
Transverse Shear Load	521.50	Kgf
Longitudinal Shear Load	1042.99	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(930/2 + 26 + 10) \sin(120/2), 2 \cdot 480)$$

$$= 867.758 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -14316 \cdot 1240 [1 - (1 - 1240/6250 + (479^2 - 0^2) / (2 \cdot 1240 \cdot 6250)) / (1 + (4 \cdot 0) / (3 \cdot 6250))]$$

$$= -3258.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 14316 \cdot 6250 / 4 ((1 + 2(479^2 - 0^2)/(6250^2)) / (1 + (4 \cdot 0) /$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 78 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9, 2026

$$(3 \cdot 6250) - 4 \cdot 1240 / 6250 \\ = 4880.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 6404 \cdot 480 / (2 \cdot 23) - 4880 / (\pi \cdot 480^2 \cdot 23) \\ = 63.88 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 6404 \cdot 480 / (2 \cdot 23) + 4880 / (\pi \cdot 480^2 \cdot 23) \\ = 69.64 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 6404 \cdot 480 / (2 \cdot 23) - 3259 / (0.11 \cdot \pi \cdot 480^2 \cdot 23) \\ = 84.80 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 6404 \cdot 480 / (2 \cdot 23) + 3259 / (0.19 \cdot \pi \cdot 480^2 \cdot 23) \\ = 56.76 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 14316 (6250 - 2 \cdot 1240) / (6250 + (4 \cdot 0 / 3)) \\ = 8635.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 8635 / (480 \cdot 23) \\ = 8.99 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{480 \cdot 23} \\ = 81.913 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(345 + 1.56 \cdot \sqrt{480 \cdot 23}, 2 \cdot 1240) \\ = 508.83 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(138 / 138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 14316 \cdot 0.1 / (509(23 + 1 \cdot 10)) \\ = -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -14316 / (4(23 + 1 \cdot 10)509) - \\ 3 \cdot 0.053 \cdot 14316 / (2(23 + 1 \cdot 10)^2) \\ = -12.31 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3720.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.12167\text{E-04} \cdot 3720 (110 - 21.1)$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 79 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

= 4.023 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	237.4	N./mm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(468 \cdot 23)} + 525)23^3)/12$$

$$= 69.641 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 69.6 + 158(163 - 11.5)^2$$

$$= 36433.320 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.8	23.0	11.5	158.0	181671.7	0.364E+05
Wearplate	525.0	10.0	28.0	52.5	147000.0	0.960E+04
Web	10.0	479.0	272.5	47.9	1305275.1	0.149E+05
BasePlate	360.0	20.0	522.0	72.0	3758400.5	0.927E+05
Totals	...	...	...	330.4	5392347.5	0.154E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5392347/330$$

$$= 163.219 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 14316$$

$$= 2913.5354 \text{ Kgf}$$

$$\text{Tension Stress, St} = (Fh / A_s) = 1.6573 \text{ N./mm}^2$$

$$\text{Allowed Stress, Sa} = 0.6 \cdot \text{Yield Str} = 142.4384 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 468 \cdot \sin(120/2) / 1.05$$

$$= 612.967 \text{ mm}$$

$$\text{Bending Moment, M} = Fh \cdot d = 1785.9365 \text{ Kg-m.}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 80 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

Bending Stress, $S_b = (M \cdot C_1 / I) = 1.8608 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 158.2649 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(14316 + 219) \cdot 360) / (4 \cdot 980 \cdot 158)}$
 $= 15.752 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (465 + 26 + 10)$
 $= 867.758 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 868 / (3 - 1)$
 $= 433.879 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 434 \cdot 360 / 2$
 $= 780.982 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 14535 / (980 \cdot 360)$
 $= 4.120 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 781 \cdot 4.12$
 $= 3217.458 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 33.5 + 21.7$
 $= 55.194 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 3217 / 55.2$
 $= 5.717 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 345 - 10$
 $= 335.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 55.2$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 81 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

= Saddle Width / 2
= 345/2
= 172.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{3422/55.2}$
= 78.739 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 1000 - $\cos(120/2) (465 + 26 + 10) - 20$
= 729.500 mm

Intermediate Term [Cc]:
= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/237)}$
= 128.941

Slenderness ratio [KL/r]:
= KL/r
= 1 • 730/78.7
= 9.265

Bending Moment [Rm]:
= $FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
= 1043 / $(2 \cdot 980) \cdot 434 \cdot 730/2$
= 84.216 Kg-m.

Compressive Allowable, $KL/r < Cc$ (9.26 < 129) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (9.26)^2 / (2 \cdot 129^2)) 237 / (5/3 + 3 \cdot (9.26) / (8 \cdot 129) - (9.26^3) / (8 \cdot 129^3))$
= 139.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= 5.72/140 + $(84.2 \cdot 172/34219688) / 158$
= 0.067

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/76.9
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 345/2
= 172.500 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 1000 - $\cos(501/2) (20 + 0 + 0) - 0$
= 479.000 mm

Radius of Gyration [r]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 82 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9, 2026

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(3425/76.9)}$$

$$= 66.747 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 479/66.7$$

$$= 7.176$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1043 / (2 \cdot 980)$$

$$= 0.532 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.53 \cdot 434 \cdot 479$$

$$= 110.595 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 111/198580$$

$$= 5.462 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.18 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.18)^2 / (2 \cdot 129^2)) 237 / (5/3 + 3 \cdot (7.18) / (8 \cdot 129) - (7.18^3) / (8 \cdot 129^3))$$

$$= 140.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 8.32/140 + (111 \cdot 172/3425) / 158$$

$$= 0.094$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	12731.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	280.4 Kg
Maximum Transverse Force	Ft	521.5 Kg
Maximum Longitudinal Force	F1	1043.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1043 / (3.04 \cdot 4)$$

$$= 8.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 521 / (3.04 \cdot 4)$$

$$= 4.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 83 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $1043 / (82.8 \cdot 4)$
= 0.3088 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1000 \cdot 521 + 0$
= 521.51 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $522 / 12732$
= 40.96 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.31, 0]
= 0.3088 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	479.50	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	246.50	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 84 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 7584.42 Kgf
Pressure used in Analysis P 6404.362 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	65.61	138.00
Long. Stress at Bottom of Midspan	67.91	138.00
Long. Stress at Top of Saddles	78.64	138.00
Long. Stress at Bottom of Saddles	60.18	138.00
Tangential Shear in Shell	4.52	110.40
Circ. Stress at Horn of Saddle	6.52	172.50
Circ. Compressive Stress in Shell	0.34	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (440/2 + 0) • 1000/868
= 761.3 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(93.7, 0) • 1000/3720
= 25.2 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(1043, 0) • 1000/3720
= 280.4 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (1043/2 + 0) • 1000/868
= 1802.9 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 5782 + max(25.2, 761, 280, 1803)
= 7584.4 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 766 (0 + $\pi/4$ (0.98 • 1.2)²)
= 919.063 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 7803.34 Kgf
Transverse Shear Load 521.50 Kgf
Longitudinal Shear Load 1042.99 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
= min(2(930/2 + 26 + 10)sin(120/2), 2•480)
= 867.758 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 85 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9, 2026

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -7584 \cdot 1240 \left[1 - \left(1 - \frac{1240}{6250} + \frac{(479^2 - 246^2)}{(2 \cdot 1240 \cdot 6250)} \right) / \left(1 + \frac{(4 \cdot 246)}{(3 \cdot 6250)} \right) \right]$$

$$= -2145.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 7584 \cdot 6250 / 4 \left(\frac{(1 + 2(479^2 - 246^2)/(6250^2))}{(1 + (4 \cdot 246)/(3 \cdot 6250))} - \frac{4 \cdot 1240}{6250} \right)$$

$$= 1951.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 6404 \cdot 480 / (2 \cdot 23) - 1951 / (\pi \cdot 480^2 \cdot 23)$$

$$= 65.61 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 6404 \cdot 480 / (2 \cdot 23) + 1951 / (\pi \cdot 480^2 \cdot 23)$$

$$= 67.91 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 6404 \cdot 480 / (2 \cdot 23) - 2145 / (0.11 \cdot \pi \cdot 480^2 \cdot 23)$$

$$= 78.64 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 6404 \cdot 480 / (2 \cdot 23) + 2145 / (0.19 \cdot \pi \cdot 480^2 \cdot 23)$$

$$= 60.18 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 7584 (6250 - 2 \cdot 1240) / (6250 + (4 \cdot 246 / 3))$$

$$= 4346.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 4346 / (480 \cdot 23)$$

$$= 4.52 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(480 \cdot 23)}$$

$$= 81.913 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(345 + 1.56 \cdot \sqrt{(480 \cdot 23)}, 2 \cdot 1240)$$

$$= 508.83 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 86 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 7584 \cdot 0.1 / (509 (23 + 1 \cdot 10)) \\
 &= -0.34 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_r, r]$:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -7584 / (4(23 + 1 \cdot 10) 509) - \\
 &\quad 3 \cdot 0.053 \cdot 7584 / (2(23 + 1 \cdot 10)^2) \\
 &= -6.52 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	237.4	N./mm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(468 \cdot 23)} + 525) 23^3) / 12 \\
 &= 69.641 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2 \\
 &= 69.6 + 158 (163 - 11.5)^2 \\
 &= 36433.320 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.8	23.0	11.5	158.0	181671.7	0.364E+05
Wearplate	525.0	10.0	28.0	52.5	147000.0	0.960E+04
Web	10.0	479.0	272.5	47.9	1305275.1	0.149E+05
BasePlate	360.0	20.0	522.0	72.0	3758400.5	0.927E+05
Totals	...	...	...	330.4	5392347.5	0.154E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 5392347 / 330 \\
 &= 163.219 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120 / 2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 7584 \\
 &= 1543.5946 \text{ Kgf}
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 87 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

Tension Stress, $St = (Fh/As) = 0.8781 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 142.4384 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$
 $= 1000 - 468 \cdot \sin(120/2)/1.05$
 $= 612.967 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 946.1913 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 0.9859 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 158.2649 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{(3(7584 + 219) \cdot 360) / (4 \cdot 980 \cdot 158)}$
 $= 11.542 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (465 + 26 + 10)$
 $= 867.758 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 868 / (3 - 1)$
 $= 433.879 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 434 \cdot 360 / 2$
 $= 780.982 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 7803 / (980 \cdot 360)$
 $= 2.212 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 781 \cdot 2.21$
 $= 1727.399 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 33.5 + 21.7$
 $= 55.194 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 1727/55.2$
 $= 3.069 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 88 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

= 345 - 10
= 335.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/55.2
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 345/2
= 172.500 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
= $\sqrt{(3422/55.2)}$
= 78.739 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1000 - cos(120/2)(465 + 26 + 10) - 20
= 729.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/237)}$
= 128.941

Slenderness ratio [KL/r]:

= KL/r
= 1 • 730/78.7
= 9.265

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 1043 / (2 • 980) • 434 • 730/2
= 84.216 Kg-m.

Compressive Allowable, KL/r < Cc (9.26 < 129) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (9.26)^2 / (2 \cdot 129^2)) 237 /$
 $(5/3 + 3 \cdot (9.26) / (8 \cdot 129) - (9.26^3) / (8 \cdot 129^3))$
= 139.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
= 3.07/140 + (84.2 • 172/34219688)/158
= 0.048

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/76.9
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 89 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9,2026

$$= 345/2$$

$$= 172.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(501/2)(20 + 0 + 0) - 0$$

$$= 479.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3425/76.9}$$

$$= 66.747 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 479/66.7$$

$$= 7.176$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1043 / (2 \cdot 980)$$

$$= 0.532 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.53 \cdot 434 \cdot 479$$

$$= 110.595 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 111/198580$$

$$= 5.462 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.18 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.18)^2 / (2 \cdot 129^2)) 237 / (5/3 + 3 \cdot (7.18) / (8 \cdot 129) - (7.18^3) / (8 \cdot 129^3))$$

$$= 140.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 4.47/140 + (111 \cdot 172/3425) / 158$$

$$= 0.066$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	6000.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	280.4 Kg
Maximum Transverse Force	Ft	521.5 Kg
Maximum Longitudinal Force	F1	1043.0 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 3819.5 \text{ Kg}$$



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FileName : 143-C

Page 90 of 172

Saddle Calcs: Operating Case: Step: 18 9:36am Sep 9, 2026

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= \max(F_f; F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3819 / (3.04 \cdot 4)$$

$$= 30.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 521 / (3.04 \cdot 4)$$

$$= 4.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 1043 / (82.8 \cdot 4)$$

$$= 0.3088 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1000 \cdot 521 + 0$$

$$= 521.51 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 522 / 6000$$

$$= 86.91 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.31, 0]$$

$$= 0.3088 \text{ cm}^2$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10, c = 3 mm, E* = 1

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 91 of 172

Saddle Calcs: Test Case:

Step: 19

9:36am

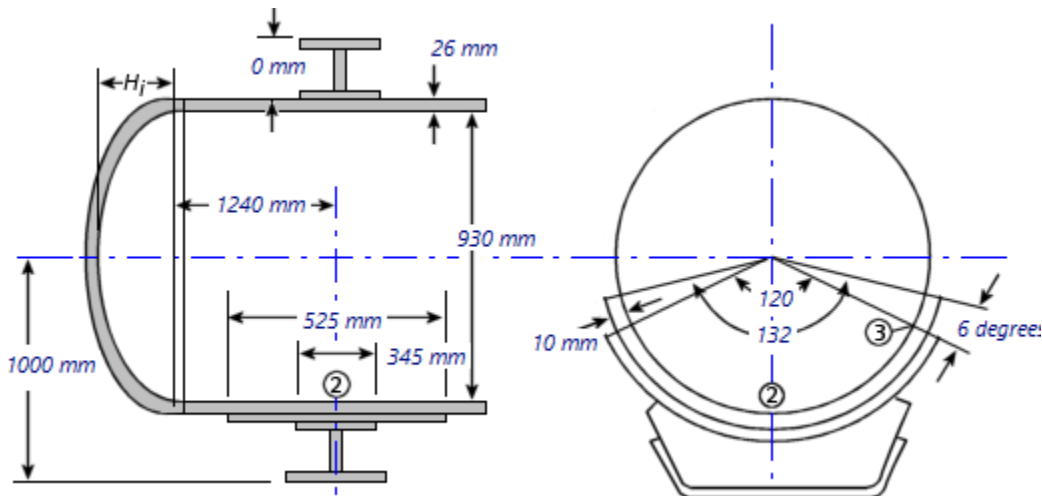
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	479.50	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		12343.40	Kgf
Pressure used in Analysis	P	8324.699	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	84.30	234.00
Long. Stress at Bottom of Midspan	89.26	234.00
Long. Stress at Top of Saddles	102.34	234.00
Long. Stress at Bottom of Saddles	78.16	234.00
Tangential Shear in Shell	7.75	138.58
Circ. Stress at Horn of Saddle	10.61	350.99
Circ. Compressive Stress in Shell	0.55	234.00



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 92 of 172

Saddle Calcs: Test Case:

Step: 19

9:36am

Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (220/2 + 0) \cdot 1000/868$$

$$= 380.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(46.9, 0) \cdot 1000/3720$$

$$= 12.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 11963 + \max(12.6, 381, 0, 0)$$

$$= 12343.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (0.98 \cdot 1.2)^2)$$

$$= 919.063 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 12562.32 Kgf

Transverse Shear Load 110.11 Kgf

Longitudinal Shear Load 46.86 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(930/2 + 26 + 10)\sin(120/2), 2 \cdot 480)$$

$$= 867.758 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0529 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923 \quad K6p = 0.0434$$

$$K7p = 0.0434$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -12343 \cdot 1240 [1 - (1 - 1240/6250 + (479^2 - 0^2)/(2 \cdot 1240 \cdot 6250))/(1 + (4 \cdot 0)/(3 \cdot 6250))]$$

$$= -2809.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 12343 \cdot 6250/4 ((1 + 2(479^2 - 0^2)/(6250^2))/(1 + (4 \cdot 0)/(3 \cdot 6250)) - 4 \cdot 1240/6250)$$

$$= 4207.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 8325 \cdot 480/(2 \cdot 23) - 4208/(\pi \cdot 480^2 \cdot 23)$$

$$= 84.30 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 93 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9, 2026

$$= 8325 \cdot 480 / (2 \cdot 23) + 4208 / (\pi \cdot 480^2 \cdot 23)$$

$$= 89.26 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 8325 \cdot 480 / (2 \cdot 23) - 2810 / (0.11 \cdot \pi \cdot 480^2 \cdot 23)$$

$$= 102.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 8325 \cdot 480 / (2 \cdot 23) + 2810 / (0.19 \cdot \pi \cdot 480^2 \cdot 23)$$

$$= 78.16 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 12343 (6250 - 2 \cdot 1240) / (6250 + (4 \cdot 0 / 3))$$

$$= 7445.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 7446 / (480 \cdot 23)$$

$$= 7.75 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(480 \cdot 23)}$$

$$= 81.913 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(345 + 1.56 \cdot \sqrt{(480 \cdot 23)}, 2 \cdot 1240)$$

$$= 508.83 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 12343 \cdot 0.1 / (509(23 + 1 \cdot 10))$$

$$= -0.55 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -12343 / (4(23 + 1 \cdot 10)509) -$$

$$3 \cdot 0.053 \cdot 12343 / (2(23 + 1 \cdot 10)^2)$$

$$= -10.61 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(468 \cdot 23)} + 525)^3) / 12$$

$$= 69.641 \text{ cm}^4$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 94 of 172

Saddle Calcs: Test Case:

Step: 19

9:36am

Sep 9, 2026

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 69.6 + 158 (163 - 11.5)^2$$

$$= 36433.320 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	686.8	23.0	11.5	158.0	181671.7	0.364E+05
Wearplate	525.0	10.0	28.0	52.5	147000.0	0.960E+04
Web	10.0	479.0	272.5	47.9	1305275.1	0.149E+05
BasePlate	360.0	20.0	522.0	72.0	3758400.5	0.927E+05
Totals	...	...	...	330.4	5392347.5	0.154E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5392347/330$$

$$= 163.219 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 12343$$

$$= 2512.1504 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.4290 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 468 \cdot \sin(120/2) / 1.05$$

$$= 612.967 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1539.8958 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.6045 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(12343 + 219) \cdot 360) / (4 \cdot 980 \cdot 173)}$$

$$= 13.994 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (465 + 26 + 10)$$

$$= 867.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 868 / (3 - 1)$$

$$= 433.879 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 95 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9,2026

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 434 \cdot 360 / 2$$

$$= 780.982 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 12562 / (980 \cdot 360)$$

$$= 3.561 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 781 \cdot 3.56$$

$$= 2780.879 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 33.5 + 21.7$$

$$= 55.194 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 2781 / 55.2$$

$$= 4.941 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 345 - 10$$

$$= 335.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 55.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 345 / 2$$

$$= 172.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (3422 / 55.2) }$$

$$= 78.739 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1000 - \cos(120 / 2) (465 + 26 + 10) - 20$$

$$= 729.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 96 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9, 2026

$$= 1 \cdot 730/78.7$$

$$= 9.265$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 46.9 / (2 \cdot 980) \cdot 434 \cdot 730/2$$

$$= 3.784 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($9.26 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.26)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (9.26) / (8 \cdot 123) - (9.26^3) / (8 \cdot 123^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.94/153 + (3.78 \cdot 172/34219688) / 173$$

$$= 0.033$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/76.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345/2$$

$$= 172.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1000 - \cos(501/2) (20 + 0 + 0) - 0$$

$$= 479.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (3425/76.9) }$$

$$= 66.747 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 479/66.7$$

$$= 7.176$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 46.9 / (2 \cdot 980)$$

$$= 0.024 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.024 \cdot 434 \cdot 479$$

$$= 4.969 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 97 of 172

Saddle Calcs: Test Case:

Step: 19

9:36am

Sep 9, 2026

$$= 4.97/198580$$

$$= 0.245 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.18 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.18)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (7.18) / (8 \cdot 123) - (7.18^3) / (8 \cdot 123^3))$$

$$= 153.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 7.19/154 + (4.97 \cdot 172/3425) / 173$$

$$= 0.048$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load QO (Weight)	QO	12181.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	12.6 Kg
Maximum Transverse Force	Ft	110.1 Kg
Maximum Longitudinal Force	F1	46.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 46.9 / (3.04 \cdot 4)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 110 / (3.04 \cdot 4)$$

$$= 0.9 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 46.9 / (29 \cdot 4)$$

$$= 0.0396 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 110 + 0$$

$$= 110.11 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 110/12182$$

$$= 9.04 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 98 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9,2026

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.04, 0]
 = 0.0396 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	479.50	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	246.50	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		5528.70	Kgf
Pressure used in Analysis	P	8324.699	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	85.94	234.00
Long. Stress at Bottom of Midspan	87.62	234.00
Long. Stress at Top of Saddles	95.44	234.00
Long. Stress at Bottom of Saddles	81.98	234.00
Tangential Shear in Shell	3.30	138.58
Circ. Stress at Horn of Saddle	4.75	350.99
Circ. Compressive Stress in Shell	0.25	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (220/2 + 0) • 1000/868
 = 380.7 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(46.9, 0) • 1000/3720
 = 12.6 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 99 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9, 2026

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 5148 + max(12.6, 381, 0, 0)
 = 5528.7 Kgf

Longitudinal Wind Force [FI]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (0.98 • 1.2)²)
 = 919.063 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5747.62	Kgf
Transverse Shear Load	110.11	Kgf
Longitudinal Shear Load	46.86	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(930/2 + 26 + 10)sin(120/2), 2•480)
 = 867.758 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -5529•1240 [1 - (1 - 1240/6250 + (479²-246²)/(2•1240•6250))/(1+(4•246)/(3•6250))]
 = -1563.6 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4 ((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 5529•6250/4 ((1+2(479²-246²)/(6250²))/(1+(4•246)/(3•6250)) - 4•1240/6250)
 = 1422.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 8325 • 480/(2•23) - 1423/(π •480²•23)
 = 85.94 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 8325 • 480/(2 • 23) + 1423/(π • 480² • 23)
 = 87.62 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 8325•480/(2•23) - 1564/(0.11• π •480²•23)
 = 95.44 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 8325•480/(2•23) + 1564/(0.19• π •480²•23)
 = 81.98 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = Q(L-2a)/(L+(4•hm/3))



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 100 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9, 2026

$$= 5529 (6250 - 2 \cdot 1240) / (6250 + (4 \cdot 246/3))$$

$$= 3168.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 3168 / (480 \cdot 23)$$

$$= 3.30 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(480 \cdot 23)}$$

$$= 81.913 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(345 + 1.56 \cdot \sqrt{(480 \cdot 23)}, 2 \cdot 1240)$$

$$= 508.83 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 5529 \cdot 0.1 / (509(23 + 1 \cdot 10))$$

$$= -0.25 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -5529 / (4(23 + 1 \cdot 10)509) -$$

$$3 \cdot 0.053 \cdot 5529 / (2(23 + 1 \cdot 10)^2)$$

$$= -4.75 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(468 \cdot 23)} + 525)23^3) / 12$$

$$= 69.641 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 69.6 + 158(163 - 11.5)^2$$

$$= 36433.320 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.8	23.0	11.5	158.0	181671.7	0.364E+05
Wearplate	525.0	10.0	28.0	52.5	147000.0	0.960E+04
Web	10.0	479.0	272.5	47.9	1305275.1	0.149E+05
BasePlate	360.0	20.0	522.0	72.0	3758400.5	0.927E+05
Totals	...	...	...	330.4	5392347.5	0.154E+06



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 101 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9, 2026

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5392347/330$$

$$= 163.219 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 5529$$

$$= 1125.2111 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.6401 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 468 \cdot \sin(120/2) / 1.05$$

$$= 612.967 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 689.7310 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.7187 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(5529 + 219) \cdot 360) / (4 \cdot 980 \cdot 173)}$$

$$= 9.465 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (465 + 26 + 10)$$

$$= 867.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 868 / (3 - 1)$$

$$= 433.879 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 434 \cdot 360/2$$

$$= 780.982 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 5748 / (980 \cdot 360)$$

$$= 1.629 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 781 \cdot 1.63$$

$$= 1272.331 \text{ Kgf}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 102 of 172

Saddle Calcs: Test Case:

Step: 19

9:36am

Sep 9, 2026

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 33.5 + 21.7

= 55.194 cm²

Compressive Stress [Sc]:

= P/Ar

= 1272/55.2

= 2.261 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 345 - 10

= 335.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/55.2

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 345/2

= 172.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(3422/55.2)}$

= 78.739 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1000 - cos(120/2)(465 + 26 + 10) - 20

= 729.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 730/78.7

= 9.265

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 46.9 / (2 • 980) • 434 • 730/2

= 3.784 Kg-m.

Compressive Allowable, KL/r < Cc (9.26 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (9.26)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (9.26) / (8 \cdot 123) - (9.26^3) / (8 \cdot 123^3))$

= 153 N./mm²



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 103 of 172

Saddle Calcs: Test Case:

Step: 19

9:36am

Sep 9, 2026

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.26/153 + (3.78 \cdot 172/34219688) / 173$$

$$= 0.016$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/76.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345/2$$

$$= 172.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(501/2) (20 + 0 + 0) - 0$$

$$= 479.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (3425/76.9) }$$

$$= 66.747 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 479/66.7$$

$$= 7.176$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 46.9 / (2 \cdot 980)$$

$$= 0.024 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.024 \cdot 434 \cdot 479$$

$$= 4.969 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4.97/198580$$

$$= 0.245 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($7.18 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.18)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (7.18) / (8 \cdot 123) - (7.18^3) / (8 \cdot 123^3))$$

$$= 153.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 3.29/154 + (4.97 \cdot 172/3425) / 173$$

$$= 0.023$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 104 of 172

Saddle Calcs: Test Case: Step: 19 9:36am Sep 9, 2026

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	5366.9 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	12.6 Kgf
Maximum Transverse Force	Ft	110.1 Kgf
Maximum Longitudinal Force	Fl	46.9 Kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 46.9 / (3.04 \cdot 4)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 110 / (3.04 \cdot 4)$$

$$= 0.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 46.9 / (82.8 \cdot 4)$$

$$= 0.0139 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 110 + 0$$

$$= 110.11 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 110 / 5367$$

$$= 20.52 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0, 0.014, 0]$$

$$= 0.0139 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 143-C

Page 105 of 172

Saddle Calcs: Ext. Pressure:

Step: 20

9:36am

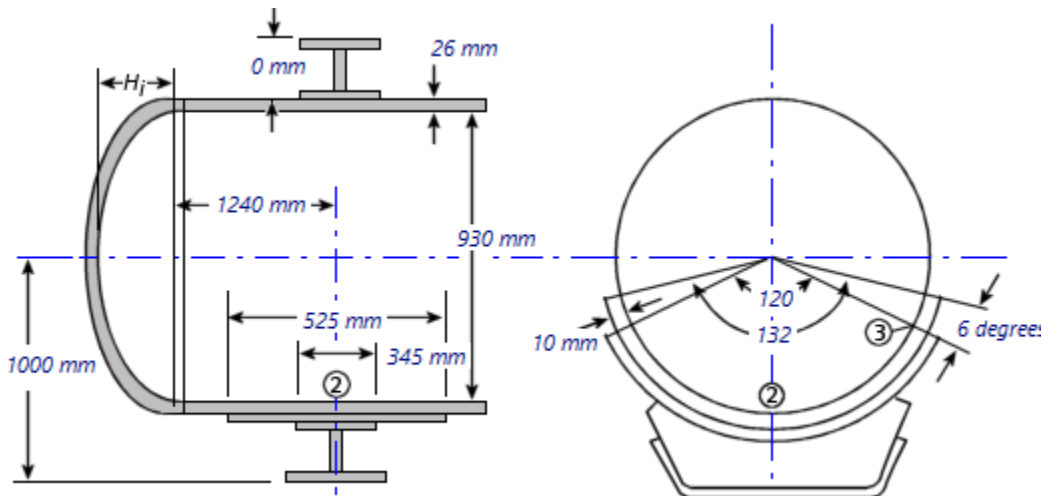
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	479.50	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	14315.60	Kgf
Pressure used in Analysis	P	-0.000 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-2.88	-213.66	
Long. Stress at Top of Midspan	2.88	138.00	
Long. Stress at Bottom of Midspan	2.88	138.00	
Long. Stress at Top of Saddles	18.04	138.00	
Long. Stress at Bottom of Saddles	-10.00	-213.66	
Long. Stress at Bottom of Saddles	10.00	138.00	
Tangential Shear in Shell	8.99	110.40	



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FileName : 143-C

Page 106 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

Circ. Stress at Horn of Saddle	12.31	172.50
Circ. Compressive Stress in Shell	0.64	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$$

$$= 3 (440/2 + 0) \cdot 1000/868$$

$$= 761.3\ Kg$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(93.7, 0) \cdot 1000/3720$$

$$= 25.2\ Kg$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(1043, 0) \cdot 1000/3720$$

$$= 280.4\ Kg$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$$

$$= 3 (1043/2 + 0) \cdot 1000/868$$

$$= 1802.9\ Kg$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 12513 + \max(25.2, 761, 280, 1803)$$

$$= 14315.6\ Kg$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(0.98 \cdot 1.2)^2)$$

$$= 919.063\ N$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	14534.52	Kgf
Transverse Shear Load	521.50	Kgf
Longitudinal Shear Load	1042.99	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(930/2 + 26 + 10) \sin(120/2), 2 \cdot 480)$$

$$= 867.758\ mm$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -14316 \cdot 1240 [1 - (1 - 1240/6250 + (479^2 - 0^2) / (2 \cdot 1240 \cdot 6250)) / (1 + (4 \cdot 0) / (3 \cdot 6250))]$$

$$= -3258.6\ Kg \cdot m$$

Moment per Equation 4.15.2 [M2]:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 107 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 14316 \cdot 6250 / 4 \cdot ((1 + 2(479^2 - 0^2) / (6250^2)) / (1 + (4 \cdot 0) / (3 \cdot 6250)) - 4 \cdot 1240 / 6250) \\
 &= 4880.2 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\
 &= -0 \cdot 480 / (2 \cdot 23) - 4880 / (\pi \cdot 480^2 \cdot 23) \\
 &= -2.88 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -0 \cdot 480 / (2 \cdot 23) + 4880 / (\pi \cdot 480^2 \cdot 23) \\
 &= 2.88 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\
 &= -0 \cdot 480 / (2 \cdot 23) - 3259 / (0.11 \cdot \pi \cdot 480^2 \cdot 23) \\
 &= 18.04 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -0 \cdot 480 / (2 \cdot 23) + 3259 / (0.19 \cdot \pi \cdot 480^2 \cdot 23) \\
 &= -10.00 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 14316 \cdot (6250 - 2 \cdot 1240) / (6250 + (4 \cdot 0 / 3)) \\
 &= 8635.2 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 8635 / (480 \cdot 23) \\
 &= 8.99 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(480 \cdot 23)} \\
 &= 81.913 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(345 + 1.56 \cdot \sqrt{(480 \cdot 23)}, 2 \cdot 1240) \\
 &= 508.83 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr / S, 1) \\
 &= \min(138 / 138, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 14316 \cdot 0.1 / (509(23 + 1 \cdot 10)) \\
 &= -0.64 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -14316 / (4(23 + 1 \cdot 10)509) - 3 \cdot 0.053 \cdot 14316 / (2(23 + 1 \cdot 10)^2) \\
 &= -12.31 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 108 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

Number of Ribs (inc. outside ribs)	Nribs	3
Rib Thickness	Ribtk	10.0000 mm
Web Thickness	Webtk	10.0000 mm
Web Location	Webloc	Center
Saddle Design Temperature		21.1 °C
Saddle Yield Stress	Sy	237.4 N./mm²
Height of Web at Center	Hw,c	500.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(468 \cdot 23)} + 525)^3) / 12$$

$$= 69.641 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 69.6 + 158 (163 - 11.5)^2$$

$$= 36433.320 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm²	AY mm³	I + AD^(2) cm⁴
Shell	686.8	23.0	11.5	158.0	181671.7	0.364E+05
Wearplate	525.0	10.0	28.0	52.5	147000.0	0.960E+04
Web	10.0	479.0	272.5	47.9	1305275.1	0.149E+05
BasePlate	360.0	20.0	522.0	72.0	3758400.5	0.927E+05
Totals	...	...	...	330.4	5392347.5	0.154E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5392347 / 330$$

$$= 163.219 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 14316$$

$$= 2913.5354 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.6573 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 142.4384 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000 - 468 \cdot \sin(120 / 2) / 1.05$$

$$= 612.967 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1785.9365 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.8608 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 158.2649 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 109 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9,2026

$$= \sqrt{3(14316 + 219)360 / (4 \cdot 980 \cdot 158)}$$

$$= 15.752 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (465 + 26 + 10)$$

$$= 867.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 868 / (3 - 1)$$

$$= 433.879 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 434 \cdot 360 / 2$$

$$= 780.982 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 14535 / (980 \cdot 360)$$

$$= 4.120 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 781 \cdot 4.12$$

$$= 3217.458 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 33.5 + 21.7$$

$$= 55.194 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 3217 / 55.2$$

$$= 5.717 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 345 - 10$$

$$= 335.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 55.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345 / 2$$

$$= 172.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 110 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

$$= \sqrt{(3422/55.2)}$$

$$= 78.739 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (465 + 26 + 10) - 20$$

$$= 729.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 237)}$$

$$= 128.941$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 730 / 78.7$$

$$= 9.265$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1043 / (2 \cdot 980) \cdot 434 \cdot 730 / 2$$

$$= 84.216 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.26 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.26)^2 / (2 \cdot 129^2)) 237 /$$

$$(5/3 + 3 \cdot (9.26) / (8 \cdot 129) - (9.26^3) / (8 \cdot 129^3))$$

$$= 139.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.72 / 140 + (84.2 \cdot 172 / 34219688) / 158$$

$$= 0.067$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 76.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345 / 2$$

$$= 172.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(501/2) (20 + 0 + 0) - 0$$

$$= 479.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3425 / 76.9)}$$

$$= 66.747 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 111 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

$$= 1 \cdot 479/66.7$$

$$= 7.176$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 1043 / (2 \cdot 980)$$

$$= 0.532 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.53 \cdot 434 \cdot 479$$

$$= 110.595 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 111/198580$$

$$= 5.462 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.18 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.18)^2 / (2 \cdot 129^2)) 237 / (5/3 + 3 \cdot (7.18) / (8 \cdot 129) - (7.18^3) / (8 \cdot 129^3))$$

$$= 140.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 8.32/140 + (111 \cdot 172/3425) / 158$$

$$= 0.094$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	479.50	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	6250.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1240.00	mm
Saddle Width	b1 or b	345.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	525.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	246.50	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		7584.42	Kgf
Pressure used in Analysis	P	-0.000	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-1.15	-213.66



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 112 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

Long. Stress at Top of Midspan	1.15	138.00
Long. Stress at Bottom of Midspan	1.15	138.00
Long. Stress at Top of Saddles	11.88	138.00
Long. Stress at Bottom of Saddles	-6.58	-213.66
Long. Stress at Bottom of Saddles	6.58	138.00
Tangential Shear in Shell	4.52	110.40
Circ. Stress at Horn of Saddle	6.52	172.50
Circ. Compressive Stress in Shell	0.34	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (440/2 + 0) \cdot 1000/868$$

$$= 761.3 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(93.7, 0) \cdot 1000/3720$$

$$= 25.2 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(1043, 0) \cdot 1000/3720$$

$$= 280.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1043/2 + 0) \cdot 1000/868$$

$$= 1802.9 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 5782 + \max(25.2, 761, 280, 1803)$$

$$= 7584.4 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (0.98 \cdot 1.2)^2)$$

$$= 919.063 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7803.34	Kgf
Transverse Shear Load	521.50	Kgf
Longitudinal Shear Load	1042.99	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(930/2 + 26 + 10)\sin(120/2), 2 \cdot 480)$$

$$= 867.758 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 113 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -7584 \cdot 1240 [1 - (1 - 1240/6250 + (479^2 - 246^2)/(2 \cdot 1240 \cdot 6250))/(1 + (4 \cdot 246)/(3 \cdot 6250))] \\
 &= -2145.1 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 7584 \cdot 6250/4 ((1 + 2(479^2 - 246^2)/(6250^2))/(1 + (4 \cdot 246)/(3 \cdot 6250)) - 4 \cdot 1240/6250) \\
 &= 1951.5 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\
 &= -0 \cdot 480/(2 \cdot 23) - 1951/(\pi \cdot 480^2 \cdot 23) \\
 &= -1.15 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= -0 \cdot 480/(2 \cdot 23) + 1951/(\pi \cdot 480^2 \cdot 23) \\
 &= 1.15 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\
 &= -0 \cdot 480/(2 \cdot 23) - 2145/(0.11 \cdot \pi \cdot 480^2 \cdot 23) \\
 &= 11.88 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -0 \cdot 480/(2 \cdot 23) + 2145/(0.19 \cdot \pi \cdot 480^2 \cdot 23) \\
 &= -6.58 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 7584 (6250 - 2 \cdot 1240)/(6250 + (4 \cdot 246/3)) \\
 &= 4346.4 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 4346/(480 \cdot 23) \\
 &= 4.52 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(480 \cdot 23)} \\
 &= 81.913 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(345 + 1.56 \cdot \sqrt{(480 \cdot 23)}, 2 \cdot 1240) \\
 &= 508.83 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$\begin{aligned}
 &= \min(Sr/S, 1) \\
 &= \min(138/138, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 7584 \cdot 0.1/(509(23 + 1 \cdot 10)) \\
 &= -0.34 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q/(4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q/(2(t + \eta \cdot tr)^2) \\
 &= -7584/(4(23 + 1 \cdot 10)509) -
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 114 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

$$3 \cdot 0.053 \cdot 7584 / (2(23 + 1 \cdot 10)^2) \\ = -6.52 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	237.4	N./mm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\ = ((1.56 \cdot \sqrt{(468 \cdot 23)} + 525)^3) / 12 \\ = 69.641 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\ = 69.6 + 158(163 - 11.5)^2 \\ = 36433.320 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	686.8	23.0	11.5	158.0	181671.7	0.364E+05
Wearplate	525.0	10.0	28.0	52.5	147000.0	0.960E+04
Web	10.0	479.0	272.5	47.9	1305275.1	0.149E+05
BasePlate	360.0	20.0	522.0	72.0	3758400.5	0.927E+05
Totals	...	...	...	330.4	5392347.5	0.154E+06

Distance to Centroid [C1]:

$$= AY / A \\ = 5392347 / 330 \\ = 163.219 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2 \\ = 180 - 120 / 2 \\ = 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ = (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ = 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.2 \cdot 7584 \\ = 1543.5946 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 0.8781 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 142.4384 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians}) \\ = 1000 - 468 \cdot \sin(120 / 2) / 1.05 \\ = 612.967 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 115 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9,2026

Bending Moment, $M = F_h \cdot d = 946.1913 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 0.9859 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 158.2649 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(7584 + 219) \cdot 360) / (4 \cdot 980 \cdot 158)}$

$= 11.542 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 120/2) (465 + 26 + 10)$

$= 867.758 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 868 / (3 - 1)$

$= 433.879 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 434 \cdot 360 / 2$

$= 780.982 \text{ cm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 7803 / (980 \cdot 360)$

$= 2.212 \text{ Kg/cm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 781 \cdot 2.21$

$= 1727.399 \text{ Kg}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 33.5 + 21.7$

$= 55.194 \text{ cm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 1727 / 55.2$

$= 3.069 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	345.0	...	34.5	...	0.342E+04

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 345 - 10$

$= 335.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0 / 55.2$

$= 0.000 \text{ mm}$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 116 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9,2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345/2$$

$$= 172.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3422/55.2}$$

$$= 78.739 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (465 + 26 + 10) - 20$$

$$= 729.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199943008/237}$$

$$= 128.941$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 730/78.7$$

$$= 9.265$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1043 / (2 \cdot 980) \cdot 434 \cdot 730/2$$

$$= 84.216 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($9.26 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.26)^2 / (2 \cdot 129^2)) 237 /$$

$$(5/3 + 3 \cdot (9.26) / (8 \cdot 129) - (9.26^3) / (8 \cdot 129^3))$$

$$= 139.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.07/140 + (84.2 \cdot 172/34219688) / 158$$

$$= 0.048$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	335.0	0.0	33.5	0.0	0.342E+04
Web	433.9	10.0	0.0	43.4	0.0	3.62
Totals	...	...	...	76.9	...	0.343E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/76.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 345/2$$

$$= 172.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(501/2) (20 + 0 + 0) - 0$$

$$= 479.000 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 117 of 172

Saddle Calcs: Ext. Pressure: Step: 20 9:36am Sep 9, 2026

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(3425/76.9)}$$

$$= 66.747 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 479/66.7$$

$$= 7.176$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 1043 / (2 \cdot 980)$$

$$= 0.532 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.53 \cdot 434 \cdot 479$$

$$= 110.595 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 111/198580$$

$$= 5.462 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.18 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.18)^2 / (2 \cdot 129^2)) 237 / (5/3 + 3 \cdot (7.18) / (8 \cdot 129) - (7.18^3) / (8 \cdot 129^3))$$

$$= 140.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 4.47/140 + (111 \cdot 172/3425) / 158$$

$$= 0.066$$

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 143-C

Page 118 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9, 2026

Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	930.00	mm
Design Length of Section	L	920.9690	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		604.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		13.7795	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	41.9393	mm
Height of Beveled Transition	L'	21.9393	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



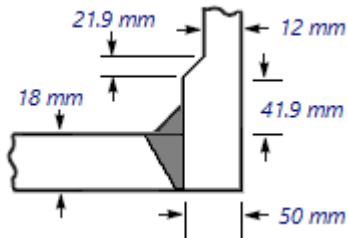
PV Elite 28 Licensee: SL User

FileName : 143-C

Page 119 of 172

Nozzle Calcs.: N1-Tube Inlet

Noz1: 9 9:36am Sep 9,2026



Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	13.780 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 468 / (138 \cdot 1 - 0.6 \cdot 1000)$
 $= 3.4064 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 178 / (138 \cdot 1 - 0.6 \cdot 1000)$
 $= 1.2956 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9278 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 38 / 0.5773$
 $= 65.8179 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	711.9986	mm
Parallel to Vessel Wall, opening length	d	355.9993	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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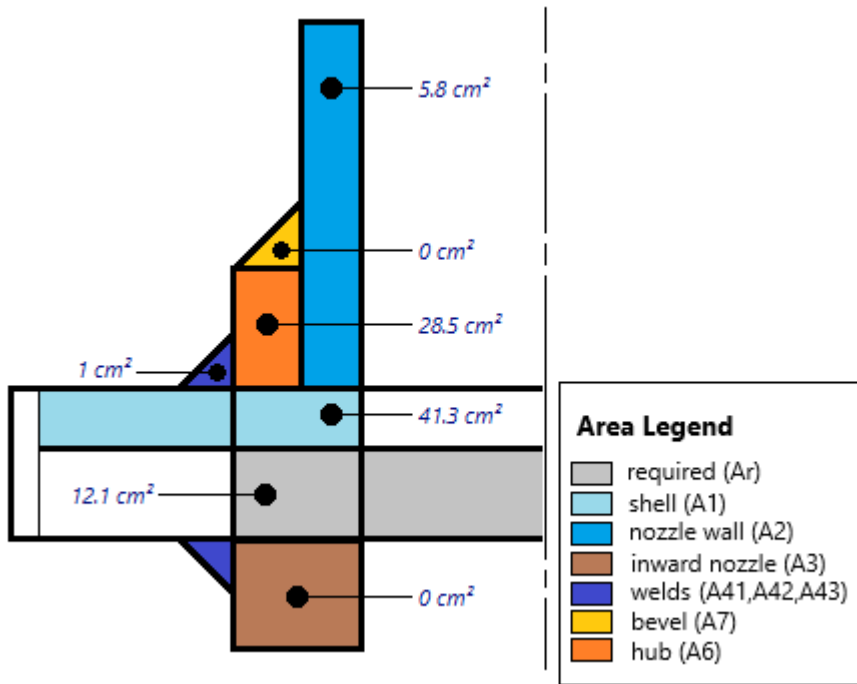
FileName : 143-C

Page 120 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.127	5.401	NA
Area in Shell	A1	41.273	42.598	NA
Area in Nozzle Wall	A2	5.778	6.054	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	28.500	28.500	NA
TOTAL AREA AVAILABLE	Atot	76.552	78.152	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (356 \cdot 3.41 \cdot 1 + 2 \cdot 47 \cdot 3.41 \cdot 1 \cdot (1 - 1))$$

$$= 12.127 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 356(1 \cdot 15 - 1 \cdot 3.41) - 2 \cdot 47$$

$$(1 \cdot 15 - 1 \cdot 3.41) \cdot (1 - 1)$$

$$= 41.273 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.5)(9 - 1.3)1$$

$$= 5.778 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 121 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_{an}/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(37.5, 200, 41.9)) \cdot (50 - 12) \cdot 1$$

$$= 28.500 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2956 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 6.4064 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 6.4064 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3509 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(6.41, 4.5)]$$

$$= 6.4064 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.3, 6.41)$$

$$= 6.4064 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33.5,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 311.5 N./mm ²	Passed
Occasional	: 9.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 18.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12, tr = 1.3, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.14, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18, tr = 3.41, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.23, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 122 of 172

Nozzle Calcs.: N1-Tube Inlet Nozl: 9 9:36am Sep 9, 2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1000.00/1960.01 = 0.510

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12.1 - 41.3 + 2 \cdot 47 \cdot 1 \cdot (1 \cdot 15 - 3.41)) \cdot 138)$
 $= \max(0, -25678.58) \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (34.3 + 1 - 0 \cdot 1) \cdot 138$
 $= 49642.12 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (34.3 + 0 + 1 + (14.1)) \cdot 138$
 $= 69483.04 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (34.3 + 0 + 1 + 0 + (14.1)) \cdot 138$
 $= 69483.04 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 450 \cdot 10 \cdot 0.49 \cdot 138$
 $= 48738. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 201) \cdot (50 - 3) \cdot 0.7 \cdot 138$
 $= 293064. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 450 \cdot (14 - 0) \cdot 0.74 \cdot 138$
 $= 103047. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (48738 + 293064) = 341802 \text{ Kgf}$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 123 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9,2026

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (48738 + 0 + 103047 + 0) = 151785 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (48738 + 103047 + 0) = 151785 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 341802 Kgf, must exceed W = 0 Kgf

or W1 = 49642 Kgf

Path 2-2 = 151784 Kgf, must exceed W = 0 Kgf

or W2 = 69483 Kgf

Path 3-3 = 151784 Kgf, must exceed W = 0 Kgf

or W3 = 69483 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3515 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 58.0600 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.0600 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	930.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	349.999	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 124 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9, 2026

*Warning - The ratio, Dn/Tn (9.000) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 450.0 + 2 \cdot 1.65 \cdot \sqrt{475.5(18.0 - 3.0)}$$

$$= 728.698 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	951.000 mm
Nozzle Outside Diameter	(d) =	449.999 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.319
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.768$$

Nr/P	=	0.054
Mr/P	=	0.085
M0/P	=	0.028
N0/P	=	0.067
MrD/Mc	=	0.258
NrDL/Mc	=	0.094
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/M1	=	0.076
NrDL/M1	=	0.049
M0D/M1	=	0.023
N0DL/M1	=	0.073

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	30	31	30	31
Outplane Stress Summary	23	30	93	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	-4	31	167	-107

Vessel Stresses (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 125 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9,2026

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			5	5	5	5
Shear Stress Summary			6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			31	37	167	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			7	7	7	7
Outplane Bending (P)			19	-19	19	-19
Outplane Membrane (Mc)			-14	-14	14	14
Outplane Bending (Mc)			-78	78	78	-78
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			15	15	15	15
Outplane Stress Summary			-50	67	136	-60

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			6	6	6	6
Inplane Bending (P)			59	-59	59	-59
Inplane Membrane (Mc)			-15	-15	15	15
Inplane Bending (Mc)			-262	262	262	-262
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			30	31	30	31
Inplane Stress Summary			-181	225	374	-267

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			5	5	5	5
Shear Stress Summary			3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			181	225	374	268



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 126 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9,2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	0	0	24	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-1	0	15	-3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	4	4	24	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	-2	-1	26	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 127 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9, 2026

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-21	21	21	-21
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-18	17	31	-20
----------------------	-----	----	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	18	18	32	48

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	-5	-5	19	19	-9	-9	21	21
Circ. Q (SUS)	-3	3	41	-41	-203	203	321	-321
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-2	-2	14	14	-7	-7	21	21
Long. Q (SUS)	-18	18	136	-136	-59	59	97	-97
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	25.1	26.1	49.0	50.0	21.1	22.1	51.1	52.1
Pm+Pl+Q (Total)	29.5	36.1	165.2	116.3	182.1	225.1	372.2	269.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	138.00	Passed
Pm+Pl (SUS)	52.06	207.00	Passed
Pm+Pl+Q (TOTAL)	372.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 128 of 172

Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 9:36am

Sep 9, 2026

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-5	-5	19	19	-7	-7	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-1	1	9	-9	-17	17	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	1.0	1.0	23.0	24.0	3.0	2.0	25.0	26.0
Pm+Pl+Q (Total)	2.6	2.0	23.1	28.1	18.1	18.1	30.2	46.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	26.00	207.00	Passed
Pm+Pl+Q (TOTAL)	46.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 129 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9,2026

Input, Nozzle Desc: N2-Tube Outlet

From: 30

Pressure for Reinforcement Calculations	P	1008.920	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	930.00	mm
Design Length of Section	L	920.9690	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		604.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		13.7795	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	41.9393	mm
Height of Beveled Transition	L'	21.9393	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 143-C

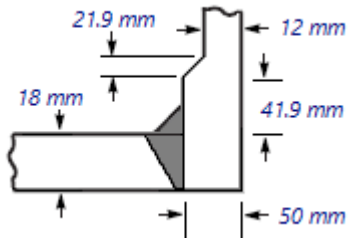
Page 130 of 172

Nozzle Calcs.: N2-Tube Outlet

Noz1: 10

9:36am

Sep 9, 2026



Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	13.780 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1009 \cdot 468 / (138 \cdot 1 - 0.6 \cdot 1009)$
 $= 3.4369 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1009 \cdot 178 / (138 \cdot 1 - 0.6 \cdot 1009)$
 $= 1.3072 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9278 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 38 / 0.5773$
 $= 65.8179 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	711.9986	mm
Parallel to Vessel Wall, opening length	d	355.9993	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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FileName : 143-C

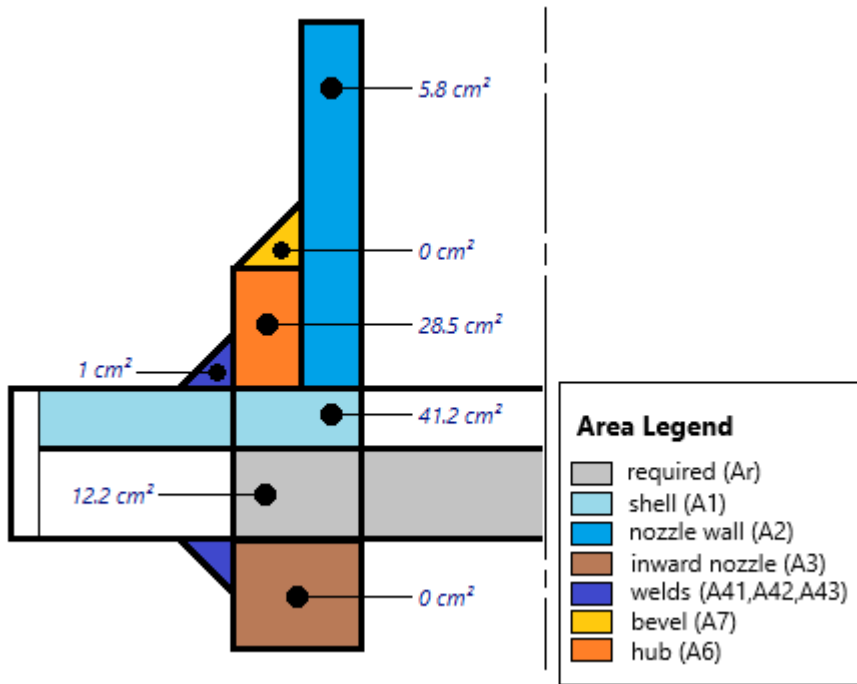
Page 131 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.235	5.401	NA
Area in Shell	A1	41.165	42.598	NA
Area in Nozzle Wall	A2	5.770	6.054	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	28.500	28.500	NA
TOTAL AREA AVAILABLE	Atot	76.434	78.152	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (356 \cdot 3.44 \cdot 1 + 2 \cdot 47 \cdot 3.44 \cdot 1 \cdot (1 - 1))$$

$$= 12.235 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 356(1 \cdot 15 - 1 \cdot 3.44) - 2 \cdot 47$$

$$(1 \cdot 15 - 1 \cdot 3.44) \cdot (1 - 1)$$

$$= 41.165 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.5)(9 - 1.31)1$$

$$= 5.770 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 132 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_{an}/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(37.5, 200, 41.9)) \cdot (50 - 12) \cdot 1$$

$$= 28.500 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3072 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 6.4369 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 6.4369 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3509 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(6.44, 4.5)]$$

$$= 6.4369 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.31, 6.44)$$

$$= 6.4369 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 311.4 N./mm ²	Passed
Occasional	: 9.7,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 18.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12, tr = 1.31, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.15, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 3.44, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.23, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 133 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10 9:36am

Sep 9, 2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1008.92/1960.01 = 0.515

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12.2 - 41.2 + 2 \cdot 47 \cdot 1 \cdot (1 \cdot 15 - 3.44)) \cdot 138)$
 $= \max(0, -25413.18) \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (34.3 + 1 - 0 \cdot 1) \cdot 138$
 $= 49629.88 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (34.3 + 0 + 1 + (14.1)) \cdot 138$
 $= 69470.79 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (34.3 + 0 + 1 + 0 + (14.1)) \cdot 138$
 $= 69470.79 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 450 \cdot 10 \cdot 0.49 \cdot 138$
 $= 48738. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 201) \cdot (50 - 3) \cdot 0.7 \cdot 138$
 $= 293064. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 450 \cdot (14 - 0) \cdot 0.74 \cdot 138$
 $= 103047. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (48738 + 293064) = 341802 \text{ Kgf}$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 134 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (48738 + 0 + 103047 + 0) = 151785 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (48738 + 103047 + 0) = 151785 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 341802 Kgf, must exceed W = 0 Kgf

or W1 = 49629 Kgf

Path 2-2 = 151784 Kgf, must exceed W = 0 Kgf

or W2 = 69470 Kgf

Path 3-3 = 151784 Kgf, must exceed W = 0 Kgf

or W3 = 69470 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3506 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 58.0600 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.0600 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	930.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	349.999	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1008.92	KPa.
Radial Load	P	-2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 135 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026

*Warning - The ratio, Dn/Tn (9.000) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 450.0 + 2 \cdot 1.65 \cdot \sqrt{475.5(18.0 - 3.0)}$$

$$= 728.698 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	951.000 mm
Nozzle Outside Diameter	(d) =	449.999 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.319
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.768$$

Nr/P	=	0.054
Mr/P	=	0.085
M0/P	=	0.028
N0/P	=	0.067
MrD/Mc	=	0.258
NrDL/Mc	=	0.094
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/M1	=	0.076
NrDL/M1	=	0.049
M0D/M1	=	0.023
N0DL/M1	=	0.073

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	30	31	30	31
Outplane Stress Summary	23	30	93	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	-4	31	167	-107

Vessel Stresses (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 136 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			5	5	5	5
Shear Stress Summary			6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			31	38	167	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			7	7	7	7
Outplane Bending (P)			19	-19	19	-19
Outplane Membrane (Mc)			-14	-14	14	14
Outplane Bending (Mc)			-78	78	78	-78
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			15	15	15	15
Outplane Stress Summary			-49	67	136	-60

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			6	6	6	6
Inplane Bending (P)			59	-59	59	-59
Inplane Membrane (Mc)			-15	-15	15	15
Inplane Bending (Mc)			-262	262	262	-262
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			30	31	30	31
Inplane Stress Summary			-181	225	375	-267

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			5	5	5	5
Shear Stress Summary			3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			181	225	375	267



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FileName : 143-C

Page 137 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	0	0	24	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-1	0	15	-3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	4	4	24	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	-2	-1	26	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 138 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-21	21	21	-21
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-18	17	31	-20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	18	18	32	48

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	-5	-5	19	19	-9	-9	21	21
Circ. Q (SUS)	-3	3	41	-41	-203	203	321	-321
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-2	-2	14	14	-7	-7	21	21
Long. Q (SUS)	-18	18	136	-136	-59	59	97	-97
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	25.1	26.1	49.0	50.0	21.1	22.1	51.1	52.1
Pm+Pl+Q (Total)	29.5	36.1	165.2	116.3	182.1	225.1	372.2	269.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	138.00	Passed
Pm+Pl (SUS)	52.06	207.00	Passed
Pm+Pl+Q (TOTAL)	372.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 139 of 172

Nozzle Calcs.: N2-Tube Outlet

Nozl: 10

9:36am

Sep 9, 2026

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-5	-5	19	19	-7	-7	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-1	1	9	-9	-17	17	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	1.0	1.0	23.0	24.0	3.0	2.0	25.0	26.0
Pm+Pl+Q (Total)	2.6	2.0	23.1	28.1	18.1	18.1	30.2	46.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	26.00	207.00	Passed
Pm+Pl+Q (TOTAL)	46.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 143-C

Page 140 of 172

Nozzle Calcs.: N3-Shell inlet

Nozl: 11

9:36am

Sep 9, 2026

Input, Nozzle Desc: N3-Shell inlet

From: 50

Pressure for Reinforcement Calculations	P	6400.000	KPa.
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	930.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3158.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		11.8110	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	46.1769	mm
Height of Beveled Transition	L'	31.1769	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 143-C

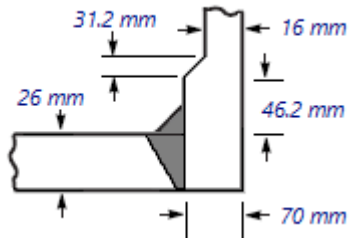
Page 141 of 172

Nozzle Calcs.: N3-Shell inlet

Nozl: 11

9:36am

Sep 9, 2026



Reinforcement CALCULATION, Description: N3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	11.811 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6400 \cdot 468 / (138 \cdot 1 - 0.6 \cdot 6400)$
 $= 22.3273 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6400 \cdot 153 / (138 \cdot 1 - 0.6 \cdot 6400)$
 $= 7.2993 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 54 / 0.5773$
 $= 93.5308 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	611.9988	mm
Parallel to Vessel Wall, opening length	d	305.9994	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	57.5000	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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FileName : 143-C

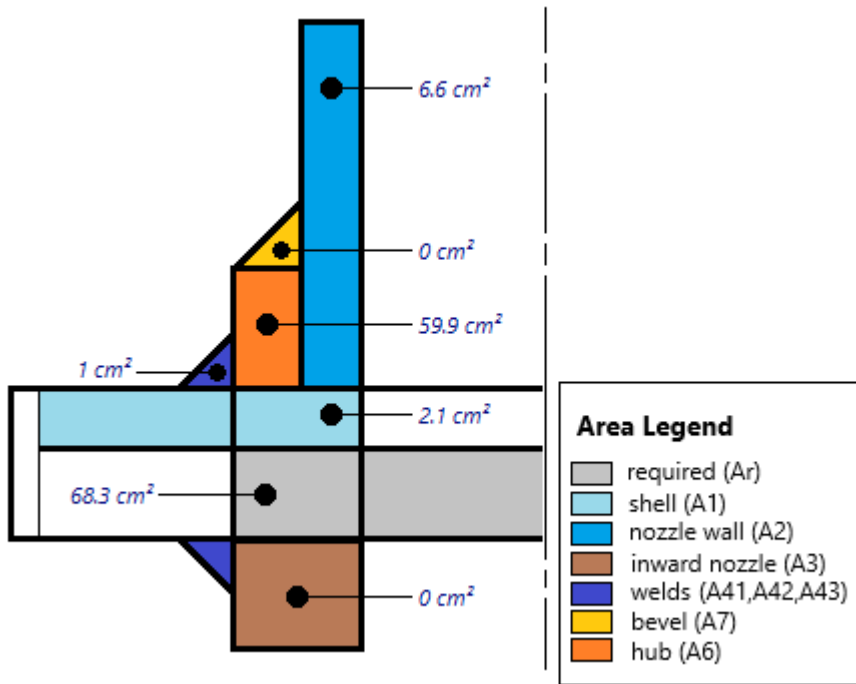
Page 142 of 172

Nozzle Calcs.: N3-Shell inlet

Nozl: 11

9:36am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	68.321	NA	NA
Area in Shell	A1	2.058	NA	NA
Area in Nozzle Wall	A2	6.556	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	59.879	NA	NA
TOTAL AREA AVAILABLE	Atot	69.493	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (306 \cdot 22.3 \cdot 1 + 2 \cdot 67 \cdot 22.3 \cdot 1 \cdot (1 - 1))$$

$$= 68.321 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 306(1 \cdot 23 - 1 \cdot 22.3) - 2 \cdot 67$$

$$(1 \cdot 23 - 1 \cdot 22.3) \cdot (1 - 1)$$

$$= 2.058 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 57.5)(13 - 7.3)1$$

$$= 6.556 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 143 of 172

Nozzle Calcs.: N3-Shell inlet

Nozl: 11

9:36am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(57.5, 300, 46.2)) \cdot (70 - 16) \cdot 1$$

$$= 59.879 \text{ cm}^2 \text{ Includes Bevel Area of } 10.008 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 10.2993 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 25.3273 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 25.3273 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(25.3, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(10.3, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 52.2,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 292.8 N./mm ²	Passed
Occasional	: 36.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 12.8,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 16, tr = 7.3, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.56, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26, tr = 22.3, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -2 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 144 of 172

Nozzle Calcs.: N3-Shell inlet Nozl: 11 9:36am Sep 9, 2026

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6400.00/10200.00 = 0.627$

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (68.3 - 2.06 + 2 \cdot 67 \cdot 1 \cdot (1 \cdot 23 - 22.3)) \cdot 138)$
 $= 94511.01 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (66.4 + 1 - 0 \cdot 1) \cdot 138$
 $= 94891.77 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (66.4 + 0 + 1 + (30.8)) \cdot 138$
 $= 138260.34 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (66.4 + 0 + 1 + 0 + (30.8)) \cdot 138$
 $= 138260.34 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 440 \cdot 10 \cdot 0.49 \cdot 138$
 $= 47655. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 186) \cdot (70 - 3) \cdot 0.7 \cdot 138$
 $= 386673. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 440 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 165529. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (47655 + 386673) = 434328 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 145 of 172

Nozzle Calcs.: N3-Shell inlet Nozl: 11 9:36am Sep 9,2026

$$= (47655 + 0 + 165529 + 0) = 213184 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (47655 + 165529 + 0) = 213184 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 434327 Kg, must exceed W = 94511 Kg or W1 = 94891 Kg

Path 2-2 = 213184 Kg, must exceed W = 94511 Kg or W2 = 138260 Kg

Path 3-3 = 213184 Kg, must exceed W = 94511 Kg or W3 = 138260 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6450 KPa.

The Drop for this Nozzle is : 55.3353 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 381.3353 mm

Input Echo, WRC297 Item 1, Description: N3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	930.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	299.999	mm
Nozzle Thickness	70.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6400.00	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.42	Kg-m.
Longitudinal Moment	Ml	1284.42	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.286) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 146 of 172

Nozzle Calcs.: N3-Shell inlet

Nozl: 11

9:36am

Sep 9, 2026

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 440.0 + 2 \cdot 1.65 \cdot \sqrt{479.5(26.0 - 3.0)}$$

$$= 786.554 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 959.000 mm

Nozzle Outside Diameter (d) = 439.999 mm

Vessel Thickness used (T) = 23.000 mm

Nozzle Thickness used (t) = 67.000 mm

T / t = 0.343

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.963$$

Nr/P = 0.060

Mr/P = 0.106

M0/P = 0.033

N0/P = 0.074

MrD/Mc = 0.265

NrDL/Mc = 0.096

M0d/Mc = 0.080

N0DL/Mc = 0.086

MrD/Ml = 0.089

NrDL/Ml = 0.056

M0D/Ml = 0.025

N0DL/Ml = 0.085

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	127	133	127	133
Outplane Stress Summary	126	131	151	124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	61	64	125	13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0



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FileName : 143-C

Page 147 of 172

Nozzle Calcs.: N3-Shell inlet Nozl: 11 9:36am Sep 9,2026

Outplane Shear (V1)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	126	131	151	124

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	44	79	105	36

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-85	85	85	-85
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127	133	127	133
Inplane Stress Summary	65	189	247	28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (V1)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress	65	189	247	38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 148 of 172

Nozzle Calcs.: N3-Shell inlet Nozl: 11 9:36am Sep 9,2026

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	22	14	22
Hoop Stress Summary	12	20	22	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	5	5	14	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	13	20	22	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	22	14	22
Hoop Stress Summary	12	20	22	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-8	8	8	-8



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 149 of 172

Nozzle Calcs.: N3-Shell inlet Nozl: 11 9:36am Sep 9,2026

Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	0	11	20	-1
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	13	20	22	31

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127	133	127	133	127	133	127	133
Circ. Pl (SUS)	-1	-1	7	7	-3	-3	7	7
Circ. Q (SUS)	1	-1	15	-15	-58	58	112	-112
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-1	-1	5	5	-1	-1	7	7
Long. Q (SUS)	-2	2	56	-56	-17	17	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	127.0	133.0	127.0	133.0	127.0	133.0	127.0	133.0
Pm+Pl (SUS)	126.0	132.0	134.0	140.0	124.0	130.0	134.0	140.0
Pm+Pl+Q (Total)	127.1	131.1	149.2	125.0	66.2	188.0	246.0	37.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133.00	138.00	Passed
Pm+Pl (SUS)	140.00	207.00	Passed
Pm+Pl+Q (TOTAL)	246.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of	Stress Values at
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 150 of 172

Nozzle Calcs.: N3-Shell inlet

Nozl: 11

9:36am

Sep 9, 2026

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	22	14	22	14	22	14	22
Circ. Pl (SUS)	-1	-1	7	7	-1	-1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	4	-4	-6	6	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	14.0	22.0	14.0	22.0	14.0	22.0	14.0	22.0
Pm+Pl (SUS)	13.0	21.0	21.0	29.0	13.0	21.0	21.0	29.0
Pm+Pl+Q (Total)	13.0	21.0	21.0	29.0	13.0	21.0	21.0	31.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	138.00	Passed
Pm+Pl (SUS)	29.00	207.00	Passed
Pm+Pl+Q (TOTAL)	31.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 143-C

Page 151 of 172

Nozzle Calcs.: N4-Shell Outlet

Nozl: 12

9:36am

Sep 9, 2026

Input, Nozzle Desc: N4-Shell Outlet

From: 50

Pressure for Reinforcement Calculations	P	6408.920	KPa.
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	930.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5358.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266	2
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		11.8110	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	46.1769	mm
Height of Beveled Transition	L'	31.1769	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 143-C

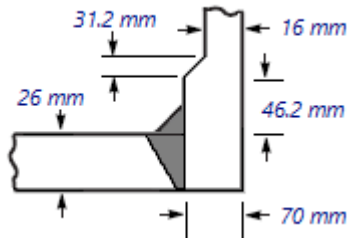
Page 152 of 172

Nozzle Calcs.: N4-Shell Outlet

Nozl: 12

9:36am

Sep 9, 2026



Reinforcement CALCULATION, Description: N4-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	11.811 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6409 \cdot 468 / (138 \cdot 1 - 0.6 \cdot 6409)$
 $= 22.3593 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6409 \cdot 153 / (138 \cdot 1 - 0.6 \cdot 6409)$
 $= 7.3098 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 54 / 0.5773$
 $= 93.5308 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	611.9988	mm
Parallel to Vessel Wall, opening length	d	305.9994	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	57.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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FileName : 143-C

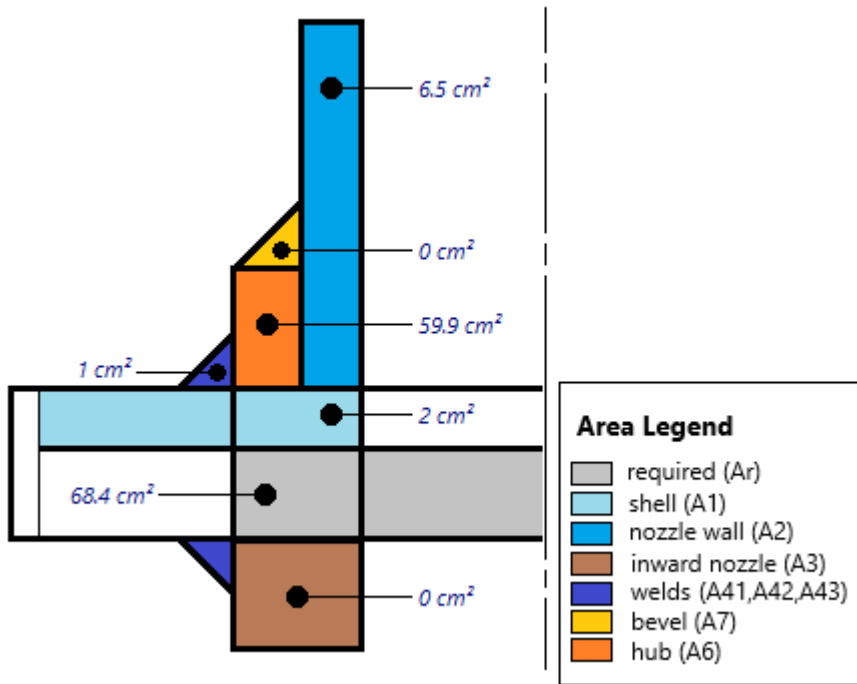
Page 153 of 172

Nozzle Calcs.: N4-Shell Outlet

Nozl: 12

9:36am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	68.419	NA	NA
Area in Shell	A1	1.960	NA	NA
Area in Nozzle Wall	A2	6.544	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	59.879	NA	NA
TOTAL AREA AVAILABLE	Atot	69.383	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (306 \cdot 22.4 \cdot 1 + 2 \cdot 67 \cdot 22.4 \cdot 1 \cdot (1 - 1))$$

$$= 68.419 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 306(1 \cdot 23 - 1 \cdot 22.4) - 2 \cdot 67$$

$$(1 \cdot 23 - 1 \cdot 22.4) \cdot (1 - 1)$$

$$= 1.960 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 57.5)(13 - 7.31)1$$

$$= 6.544 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 154 of 172

Nozzle Calcs.: N4-Shell Outlet

Nozl: 12

9:36am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(57.5, 300, 46.2)) \cdot (70 - 16) \cdot 1$$

$$= 59.879 \text{ cm}^2 \text{ Includes Bevel Area of } 10.008 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 10.3098 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 25.3593 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 25.3593 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(25.4, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(10.3, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 65.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 279.4 N./mm ²	Passed
Occasional	: 36.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 19.5,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 16, tr = 7.31, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.56, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26, tr = 22.4, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C



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FileName : 143-C

Page 155 of 172

Nozzle Calcs.: N4-Shell Outlet Nozl: 12 9:36am Sep 9,2026

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6408.92/10200.00 = 0.628$

Weld Size Calculations, Description: N4-Shell Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (68.4 - 1.96 + 2 \cdot 67 \cdot 1 \cdot (1 \cdot 23 - 22.4)) \cdot 138)$
 $= 94726.34 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (66.4 + 1 - 0 \cdot 1) \cdot 138$
 $= 94874.84 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (66.4 + 0 + 1 + (30.8)) \cdot 138$
 $= 138243.41 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (66.4 + 0 + 1 + 0 + (30.8)) \cdot 138$
 $= 138243.41 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 440 \cdot 10 \cdot 0.49 \cdot 138$
 $= 47655. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 186) \cdot (70 - 3) \cdot 0.7 \cdot 138$
 $= 386673. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 440 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 165529. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (47655 + 386673) = 434328 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 156 of 172

Nozzle Calcs.: N4-Shell Outlet

Nozl: 12

9:36am

Sep 9, 2026

$$= (47655 + 0 + 165529 + 0) = 213184 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (47655 + 165529 + 0) = 213184 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 434327 Kg, must exceed W = 94726 Kg or W1 = 94874 Kg

Path 2-2 = 213184 Kg, must exceed W = 94726 Kg or W2 = 138243 Kg

Path 3-3 = 213184 Kg, must exceed W = 94726 Kg or W3 = 138243 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6441 KPa.

The Drop for this Nozzle is : 55.3353 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 381.3353 mm

Input Echo, WRC297 Item 1, Description: N4-Shell Outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	930.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	299.999	mm
Nozzle Thickness	70.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6408.92	KPa.
Radial Load	P	-3212.32	Kgf
Circumferential Shear	Vc	2447.48	Kgf
Longitudinal Shear	Vl	2447.48	Kgf
Circumferential Moment	Mc	2293.62	Kg-m.
Longitudinal Moment	Ml	2293.62	Kg-m.
Torsional Moment	Mt	2752.36	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.286) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 157 of 172

Nozzle Calcs.: N4-Shell Outlet

Nozl: 12

9:36am

Sep 9, 2026

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 440.0 + 2 \cdot 1.65 \cdot \sqrt{479.5(26.0 - 3.0)}$$

$$= 786.554 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 959.000 mm

Nozzle Outside Diameter (d) = 439.999 mm

Vessel Thickness used (T) = 23.000 mm

Nozzle Thickness used (t) = 67.000 mm

T / t = 0.343

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.963$$

Nr/P = 0.060

Mr/P = 0.106

M0/P = 0.033

N0/P = 0.074

MrD/Mc = 0.265

NrDL/Mc = 0.096

M0d/Mc = 0.080

N0DL/Mc = 0.086

MrD/Ml = 0.089

NrDL/Ml = 0.056

M0D/Ml = 0.025

N0DL/Ml = 0.085

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	127	133	127	133
Outplane Stress Summary	121	132	166	120

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	47	75	162	-17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 158 of 172

Nozzle Calcs.: N4-Shell Outlet Nozl: 12 9:36am Sep 9,2026

Outplane Shear (V1)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	121	132	167	137

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-46	46	46	-46
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	25	93	134	18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-153	153	153	-153
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127	133	127	133
Inplane Stress Summary	6	243	331	-44

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (V1)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress	25	243	331	64

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 159 of 172

Nozzle Calcs.: N4-Shell Outlet Nozl: 12 9:36am Sep 9,2026

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	22	14	22
Hoop Stress Summary	10	18	26	34

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	3	5	18	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	11	18	27	34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	22	14	22
Hoop Stress Summary	10	18	27	34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-14	14	14	-14



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 160 of 172

Nozzle Calcs.: N4-Shell Outlet Nozl: 12 9:36am Sep 9,2026

Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-7	15	29	-6
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	18	18	30	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127	133	127	133	127	133	127	133
Circ. Pl (SUS)	-4	-4	12	12	-6	-6	12	12
Circ. Q (SUS)	-3	3	25	-25	-116	116	190	-190
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-2	-2	8	8	-4	-4	12	12
Long. Q (SUS)	-14	14	88	-88	-35	35	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	127.0	133.0	127.0	133.0	127.0	133.0	127.0	133.0
Pm+Pl (SUS)	123.0	129.0	139.0	145.0	121.0	127.0	139.0	145.0
Pm+Pl+Q (Total)	120.2	132.3	164.7	137.1	24.2	243.0	329.1	63.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133.00	138.00	Passed
Pm+Pl (SUS)	145.01	207.00	Passed
Pm+Pl+Q (TOTAL)	329.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of	Stress Values at
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 161 of 172

Nozzle Calcs.: N4-Shell Outlet

Nozl: 12

9:36am

Sep 9, 2026

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	22	14	22	14	22	14	22
Circ. Pl (SUS)	-4	-4	12	12	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-1	1	7	-7	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	14.0	22.0	14.0	22.0	14.0	22.0	14.0	22.0
Pm+Pl (SUS)	10.0	18.0	26.0	34.0	10.0	18.0	26.0	34.0
Pm+Pl+Q (Total)	10.1	18.1	26.1	34.0	17.1	18.3	27.6	41.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	138.00	Passed
Pm+Pl (SUS)	34.00	207.00	Passed
Pm+Pl+Q (TOTAL)	41.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 162 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: U Tube

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	6400.00 KPa.
Shell Maximum Operating Pressure	Psox,max	6400.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	26.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	930.000 mm
Shell Temperature for Internal Pressure	Ts	110.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	930.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	482
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5650.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5496.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	165.57 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	439.916 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 163 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9, 2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	4723.3 cm ²
Perimeter of Tube Layout	Cp	2662.7 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

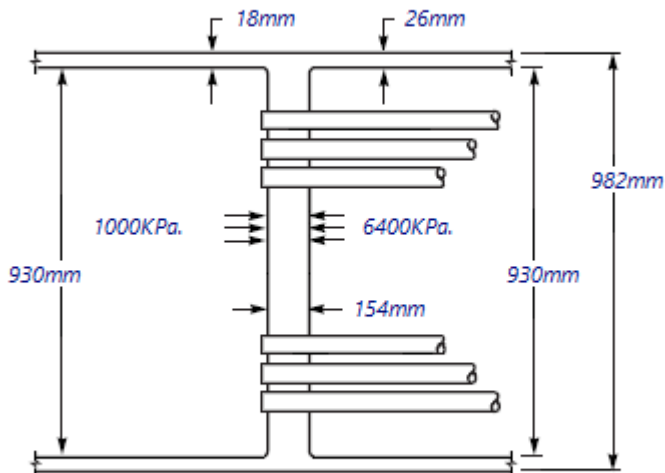
Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	154.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	982.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
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Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Channel Side Min. Design Pressure	Ptd,min 103.4190 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 164 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9,2026

Elastic Mod. at Design Temperature	110.0 °C	0.19753E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 1000)$
 $= 0.1369 + 0.0000 = 0.1369 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 82.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5935.92	12.04	50.0000	0.0075908	95.82

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.8) / (3 \cdot 12) = 10612 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.3430	25.40	5935.92	18.91	50.0000	0.0030751	92.27

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 92.3) / (3 \cdot 18.9) = 6504 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.3430 mm
Allowable Internal Pressure at Corroded thickness	16443.08 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	10612.12 KPa.
Required External Design Pressure	6503.42 KPa.
Required Thickness due to Shell Side pressure	1.3430 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot r_o + d_t$
 $= 2 \cdot 440 + 25.4$
 $= 905.232 \text{ mm}$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 165 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9, 2026

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 930 / 905$$

$$= 1.0274$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 930 / 905$$

$$= 1.0274$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 154$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 2.11 (199214080 / 199214080) \cdot (92.4 / 138) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 905 \cdot 31.8) / (\pi \cdot 905^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 4.850394$; $\mu^* = 0.200000$; $E^* = 30947736 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (4.850) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (930 + 26) 26 }$$

$$= 0.0115 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.012 \cdot 197530928 \cdot 26^3 / (6 (1 - 0.3^2))$$

$$= 7331663360.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(930 + 26) \cdot 7.33E+09 / 154^3 \cdot (1 + 154 \cdot 0.012 + (154 \cdot 0.012)^2 / 2)$$

$$= 50113396.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 930^2 / (4 \cdot 197530928 \cdot 26) \cdot (1 - 930 \cdot 0.3 / (2(930 + 26)))$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 166 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9, 2026

$$= 0.35956\text{E-}01 \text{ mm/N./mm}^2$$

Parameter [ws]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ &= 1.03 \cdot 7.33\text{E}+09 \cdot 0.012 \cdot 0.035955936 (1 + 154 \cdot 0.012) \\ &= 8668.1875 \text{ mm}^2 \end{aligned}$$

Channel coefficient [βc]:

$$\begin{aligned} &= \left(\left(12(1 - v_c^2) \right)^{1/4} \right) / \sqrt{\left((D_c + t_c) t_c \right)} \\ &= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \sqrt{\left((930 + 18) 18 \right)} \\ &= 0.0139 \text{ 1/mm} \end{aligned}$$

Channel coefficient [kc]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2)) \\ &= 0.014 \cdot 199214080 \cdot 18^3 / (6(1 - 0.3^2)) \\ &= 2961148416.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$\begin{aligned} &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ &= 6(930 + 18 / \cos(0)) \cdot 2.96\text{E}+09 / 154^3 \cdot (1 + 154 \cdot 0.014 \cdot (1 + 1) / 2 + (154 \cdot 0.014)^2 / 2 \cdot 1) \\ &= 25084858.0 \text{ KPa} \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned} &= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c))) \\ &= 930^2 / (4 \cdot 199214080 \cdot 18) \cdot (1 - 930 \cdot 0.3 / (2(930 + 18))) \\ &= 0.51423\text{E-}01 \text{ mm/N./mm}^2 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ &= 1.03 \cdot 2.96\text{E}+09 [0.051423285 (1 + 0.014 \cdot 1 \cdot 154) - 0.000 (1/154 + 1 \cdot 0.014/2)] \\ &= 6842.8525 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 982/905 \\ &= 1.0848 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 30947736 \cdot (50113396 + 25084858 + 199214080 \cdot \ln(1.08)) \\ &= 1.5993 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\ &= 905^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 6400 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 1000] \\ &= 15553829.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 167 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9, 2026

$$= 15553829 + 6843 \cdot 1000 - 8668 \cdot 6400$$

$$= -33079720.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((-33079720) - 905^2/32 \cdot 1.6 (6400 - 1000)) / (1 + 1.6)$$

$$= -97808064.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= -97808064 + 905^2/64 \cdot (3 + 0.46) \cdot (6400 - 1000)$$

$$= 141319872.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| -97808064 | ; | 141319872 |)$$

$$= 141319872.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 141319872 / (0.2 (154 - 0)^2)$$

$$= 178.7756 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 134001536 / (0.2 (121 - 0)^2)$$

$$= 275.7979 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 121 + 0 + 0$$

$$= 120.7349 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB} ; H_{reqS})$$

$$= \max(121 ; 43.4)$$

$$= 120.7349 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 6400 - 1000 |$$

$$= 5400.000 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 154 / 905 \cdot \min(0.8 \cdot 138 ; 0.533 \cdot 230)$$

$$= 15.025 \text{ N./mm}^2$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 4723 / 2663) \cdot | 6400 - 1000 | + 0$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 168 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9, 2026

= 43.3858 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 26.000mm

Min. Shell len. adjacent to the tubesheet is: 279.899 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 930^2 / (4 \cdot 26 (930 + 26)) \cdot 6400$$

$$= 55.6775 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 1648261 / 26^2 \cdot [0.012 \cdot 0.23 + 6(1 - 0.46) / 30947736 \cdot$$

$$905 / 154^3 \cdot (1 + 154 \cdot 0.012 / 2) \cdot (-97808064 + 905^2 / 32 \cdot$$

$$(6400 - 1000))]$$

$$= 302.050 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 357.727 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Note: For a given Channel thickness of: 18.000 mm

Min. Channel len. adjacent to the tubesheet is: 232.890mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 930^2 / (4 \cdot 18 (930 + 18) \cdot \cos(0)) \cdot 1000$$

$$= 12.672 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 665708 / 18^2 \cdot [(0.051423285 \cdot 1 \cdot 0.014 - 0.000 \cdot 1) \cdot 1000 -$$

$$6((1 - 0.46) / 30947736) \cdot 905 / 154^3 \cdot (1 +$$

$$0.014 \cdot 154 / 2 \cdot 1) \cdot (-97808064 + 905^2 / 32 \cdot (6400 - 1000))]$$

$$= -80.377 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 93.049 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$$

$$= 197530928 \cdot \sqrt{(1.5 \cdot 138 / 358)}$$

$$= 150258928.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 144873728 / (0.2 (154 - 0)^2)$$

$$= 183.3 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 169 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9, 2026

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 123 + 0 + 0 = 123 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}, H_{reqS}) = \max(123, 43.4) = 123 \text{ mm}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 2.11) 2.11 \\ &= 1.544 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (25.4 - 2.11) 92.4 \\ &= 1454.579 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.4 \\ &= 1423.4542 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 92.4 / 92.4 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.4 \\ &= 921.0587 \text{ Kg} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\ &= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right) \\ &= 2.667 \text{ mm} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1455 is $\geq$ tube strength 1455 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	183.3	$\leq 276.0 \text{ N./mm}^2$	Ok
Tubesheet shear stress	31.1	$\leq 110.4 \text{ N./mm}^2$	Ok
Stress in Shell at Tubesheet	357.7	$\leq 474.8 \text{ N./mm}^2$	Ok
Stress in Channel at Tubesheet	93.0	$\leq 207.0 \text{ N./mm}^2$	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
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PV Elite 28 Licensee: SL User

FileName : 143-C

Page 170 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9,2026

Tubesheet Thickness :	122.904	154.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case	Pass/	
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	51.027	...	34	276	6	110	Ps+Pt D1	Ok
D2uc	136.929	...	223	276	37	110	Ps+Pt D2	Ok
D3uc	122.904	...	183	276	31	110	Ps+Pt D3	Ok
D4uc	16.933	...	4	276	1	110	Ps+Pt D4	Ok
D1c	56.432	...	39	276	6	110	Ps+Pt-c D1	Ok
D2c	148.132	...	256	276	39	110	Ps+Pt-c D2	Ok
D3c	133.661	...	211	276	32	110	Ps+Pt-c D3	Ok
D4c	22.917	...	4	276	1	110	Ps+Pt-c D4	Ok

Max: 148.1322 ... mm 0.928 0.353 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	27	207	77	207	Ok
D2uc	...	...	387	475	152	207	Ok
D3uc	...	...	358	475	93	207	Ok
D4uc	...	...	3	207	8	207	Ok
D1c	...	...	34	207	93	207	Ok
D2c	...	...	461	475	190	207	Ok
D3c	...	...	424	475	117	207	Ok
D4c	...	...	3	207	10	207	Ok

Max ... 0.971 0.916

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	148.132	154.000	Ok
Tube thickness :	1.343	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Given Shell Thickness: 26.0000 mm

Given Channel Thickness: 18.0000 mm



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 171 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9,2026

Min Shell length of thk, (26.000) adj. to tubesheet: 279.899 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 232.890 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7069.93	1.000	7274.70	1.000
Tubesheet Shear Stress	18420.75	1.000	18420.75	1.000
Shell Stress (Axial, Junction)	6339.92	0.448	6638.88	1.000
Tubesheet-Channel Junction	5189.83	1.000	6638.88	0.948
Tube Pressure Stress	16443.07	1.000	10612.12	1.000
Minimum MAWP	5189.83		6638.88	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	8086.74	1.000	7992.83	1.000
Tubesheet Shear Stress	19167.54	1.000	19167.53	1.000
Shell Stress (Axial, Junction)	19167.54	0.987	7992.83	1.000
Tubesheet-Channel Junction	6800.54	1.000	7992.83	1.000
Tube Pressure Stress	16443.07	1.000	10612.12	1.000
Minimum MAPnc	6800.54		7992.83	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(154/4, \min(154, 26)) \\
 &= 38.500 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(6400/7993, 1) \\
 &= 0.801
 \end{aligned}$$

Thickness Ratio = 0.8, Temperature Reduction per Fig. UCS 66.1 = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(154/4, \min(154, 18)) \\
 &= 38.500 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$



PV Elite 28 Licensee: SL User

FileName : 143-C

Page 172 of 172

ASME TS Calc:

Case: 1 9:36a Sep 9, 2026

= min(1000/6801, 1)
= 0.147

Thickness Ratio = 0.15, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: